



New transshipment type MINLP model for heat exchanger network synthesis



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HIGHLIGHTS

- A novel MINLP model for simultaneous design of HEN is presented.
- Non-isothermal mixing, stream by-pass, and recycling flows are all included while keeping all constraints linear.
- A more accurate formulation for heat exchanger area is proposed by piecewise calculation.
- Better results are obtained for several literature examples.

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ABSTRACT

Heat exchanger network (HEN) is one of the most important parts in chemical process industry, owing to its significant advantages of energy recovery. However, models with plentiful HEN structures always result in complex mixed-integer nonlinear programming (MINLP) models with non-convexity of nonlinear constraints, which will cause heavy computational burdens. The computational time of MINLP models and even the feasibility of models without good initial points are still challenging researchers. This paper presents a novel transshipment type model for heat exchanger network synthesis (HENS), which is formulated as a MINLP problem with all linear constraints. Stream splitting, stream by-pass, isothermal and non-isothermal mixing, and recycling flows are all included, while multiple utilities are available in the model. Furthermore, new formulations are presented to calculate the temperature difference of heat exchangers, while a more precise heat exchanger area calculation method by piecewise calculation is developed in the new model. Six literature examples are presented to illustrate the effectiveness and applicability of the proposed model. It is shown that several better results are obtained by the proposed model.

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1. Introduction

Process industry is an energy intensive industry. Owing to the rising prices of fossil fuels as well as more and more strict environmental regulations, process industry is looking for effective ways of improving their energy efficiency. Over the past four decades, heat integration which is related to heat exchanger network synthesis (HENS) is recognized as an effective way of energy recovery in both practice and research. Many methodologies have been developed for the optimal synthesis of heat exchanger network (HEN). Comprehensive reviews about HENS can be found in Gundersen and Naess (1988), Jezowski (1994a, b), and Furman and Sahinidis (2002).

The methodologies of HEN synthesis can be classified into two types: pinch technology and mathematical programming methodologies. Pinch technology is essentially a heuristic design approach

derived from physical insights. Umeda et al. (1978) and Linnhoff and Flower (1978a, b) firstly identified the heat recovery pinch point. Linnhoff and Hindmarsh (1983) proposed the pinch design method (PDM). The design problem was divided into subproblems by pinch point. Subsequently, several work (Ahmad et al., 1990; Linnhoff and Ahmad, 1990; Townsend, 1984) based on pinch technology was presented to target the heat exchanger surface area, the energy, and the capital cost. Recently, a T-Q diagram method (Kang et al., 2016) based on the partition and merging of intervals was proposed for large-scale heat exchanger network synthesis.

Mathematical programming methods for HENS include sequential methods and simultaneous methods. Sequential synthesis methods divide the HENS problem into several subproblems, which generally include the minimum utility usage, the minimum number of exchanger units, and the minimum area cost of the network. Cerda and Westerburg (1983) formulated a linear programming (LP) problem to obtain the utility target using the transportation problem formulations. With the utility target, a

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Nomenclature

Subscript

e	heat exchanger including heater and cooler
k, k'	temperature interval
l, l'	hot stream
ls, ls'	hot stream substream
m, m'	cold stream
ms, ms'	cold stream substream
n, n'	temperature level

Sets

C	cold stream, $H=CP \cup CU$, $C=(m m=1,2,3 \dots M)$
C_k	cold stream in temperature interval k , $C_k \subset C$
CMS	cold substream, $CLS=((m,ms) m \in C, ms \in MS, (m,ms)=(1,1), (2,1) \dots)$
EX	heat exchangers between a hot process substream and a cold process substream, $EX=(e e=1,2,3 \dots E)$
H	hot stream, $H=HP \cup HU$, $H=(l l=1,2,3 \dots L)$
H_k	hot stream in temperature interval k , $H_k \subset H$
HU	hot utility, $HU \subset H$
HP	hot process stream, $HP \subset H$
HLS	hot substream, $HLS=((l,ls) l \in H, ls \in LS, (l,ls)=(1,1), (2,1) \dots)$
LS	substream of hot stream, $LS=(ls ls=1,2,3 \dots)$
MS	substream of cold stream, $MS=(ms ms=1,2,3 \dots)$
TL	temperature level, $TL=(n n=1,2,3 \dots N)$, $N=K+1$
TI	temperature interval, $TI=(k k=1,2,3 \dots K)$

Positive variables

cu_m	cold utility m consumption, kW
CUC	cold utility cost, \$
$dt_{l,ls,m,ms,k}$	temperature difference of heat transfer match, K
$dthu_m$	temperature difference of heat transfer match for hot utility L , K
$dtcu_l$	temperature difference of heat transfer match for cold utility M , K
EXC	exchanger cost, \$
EXS	sum of heat exchangers including heaters and coolers
$fl_{l,ls',n',ls,n}$	recycling mass flow from l,ls',n' to l,ls,n , kg/s
$flsk_{l,ls,k}$	flowrate of hot substream l,ls in temperature interval k , kg/s
$fmr_{m,ms',n',ms,n}$	recycling mass flow from m,ms',n' to m,ms,n , kg/s
$fmsk_{m,ms,k}$	flowrate of cold substream m,ms in temperature interval k , kg/s
hu_l	hot utility l consumption, kW
HUC	hot utility cost, \$
$q_{l,ls,m,ms,k}$	heat load of heat exchanger between hot stream substream (l,ls) and cold stream substream (m,ms) in temperature interval k , kW
$qq_{l,ls,m,ms,k,e}$	heat load of heat exchanger e between hot stream substream (l,ls) and cold stream substream (m,ms) in temperature interval k , kW
$rqhs_{l,ls,n}$	residual energy of hot process substream l,ls in temperature level n , kW
$rqhs'_{l,ls,n}$	residual energy of hot process substream l,ls in temperature level n , kW
$rqcs_{m,ms,n}$	residual energy of cold process substream m,ms in temperature level n , kW
$rqcs'_{m,ms,n}$	residual energy of cold process substream m,ms in temperature level n , kW
TAC	total annual cost, \$
$til_{l,ls,k}$	inlet temperature of hot substream l,ls in temperature interval k , K

$tim_{m,ms,k}$	inlet temperature of cold substream m,ms in temperature interval k , K
$tol_{l,ls,k}$	outlet temperature of hot substream l,ls in temperature interval k , K
$tom_{m,ms,k}$	outlet temperature of cold substream m,ms in temperature interval k , K

Binary variable

$mlsk_{l,ls,n}$	existence of exporting mass flow from temperature level n of hot substream l,ls
$mmsk_{m,ms,n}$	existence of exporting mass flow from temperature level n of cold substream m,ms
zzc_l	existence of a cooler for hot stream l in the last temperature interval K
zzh_m	existence of a heater for cold stream m in the first temperature interval 1
$z_{l,ls,m,ms,k}$	existence of a heat transfer match between hot substream l,ls and cold substream m,ms in temperature interval k
$zlsk_{l,ls,n}$	existence of importing mass flow to temperature level n of hot substream l,ls
$zmsk_{m,ms,n}$	existence of importing mass flow to temperature level n of cold substream m,ms
$zz_{l,ls,m,ms,k}$	existence of the heat exchanger between hot substream l,ls and cold substream m,ms in temperature interval k
$ZZS_{l,ls,m,ms,k}$	sum of $zz_{l,ls,m,ms,k}$ from $k=1$ to k
$x_{l,ls,m,ms,k,e}$	indicator to estimate whether $ZZS_{l,ls,m,ms,k}$ is equal to e

Parameters

A	annualized area cost coefficient for heat exchanger
B	area cost exponent for heat exchanger
CCU	per unit cost for cold utility, \$/(kW yr)
CHU	per unit cost for hot utility, \$/(kW yr)
CL_m	total heat load of cold process stream m , kW
$CLK_{m,k}$	available heat load of cold stream m in temperature interval k , kW
c_p	heat capacity of water, kJ/(kg K)
E	maximum number of heat exchangers between hot process substream l,ls and cold process substream m,ms
FC	fixed charge for heat exchanger, \$
fl_l	flowrate of hot process stream l , kg/s
fm_m	flowrate of cold process stream m , kg/s
HL_l	total heat load of hot process stream l , kW
$HLK_{l,k}$	available heat load of hot stream l in temperature interval k , kW
$MRL_{l,n}$	maximum residual energy of hot stream l in temperature level n , kW
$MRM_{m,n}$	maximum residual energy of cold stream m in temperature level n , kW
ne_l	ending temperature level of hot stream l , $ne_l \in TL$
ne_m	starting temperature level of cold stream m , $ne_m \in TL$
ns_l	starting temperature level of hot stream l , $ns_l \in TL$
ns_m	starting temperature level of cold stream m , $ns_m \in TL$
tce_m	ending temperature of cold stream m , K
tcs_m	starting temperature of cold stream m , K
the_l	ending temperature of hot stream l , K
ths_l	starting temperature of hot stream l , K
tn_n	temperature of temperature level n , K
$U_{l,m}$	heat transfer coefficient, kW/(m ² K)
$\Omega_{l,m,k}$	upper bound for heat load of heat transfer matches between hot stream l and cold stream m in temperature interval k , kW

Abbreviations

$\Delta T_{\min}$	minimum temperature difference
CUC	cold utility cost
EXC	exchanger cost
HEN	heat exchanger network
HENS	heat exchanger network synthesis
HENs	heat exchanger networks

HUC	hot utility cost
LP	linear programming
MILP	mixed-integer linear programming
MINLP	mixed-integer nonlinear programming
NLP	nonlinear programming
TAC	total annual cost
UC	utility cost

mixed integer linear programming (MILP) formulation was solved to determine the minimum number of heat transfer matches. Besides, Papoulias and Grossmann (1983) proposed several formulations based on the transshipment model. The LP and MILP versions were used for obtaining the minimum utility cost and the minimum number of units respectively. Subsequently, a vertical heat transfer model was proposed by Gundersen and Grossmann (1990), based on the idea of selecting matches to transfer heat vertically between composite curves to the largest extent. Then, an extended model (Gundersen et al., 1996) was developed to select heat load distributions that give the networks with low total area and low capital cost. Besides, Barbaro and Bagajewicz (2005, 2006) proposed a MILP formulation to give cost-optimal solutions in one step, based on the transportation/transshipment concepts.

In contrast to sequential methods, simultaneous methods use a one-step mathematical programming formulations without any decomposition. The trade-offs between various factors that affect the total annual cost are considered holistically. Floudas and Ciric (1989) proposed a match-network hyperstructure to determine the minimum cost of heat exchangers, by combining the transshipment model of Papoulias and Grossmann (1983) and the network topology of Floudas et al. (1986). In subsequent work, Ciric and Floudas extended the model to the pseudo-pinch problems (Ciric and Floudas, 1990) and to include the optimization of the minimum utility cost (Ciric and Floudas, 1991). Recently, Kim and Bagajewicz (2016) presented an extension of the hyperstructure of Floudas et al. (1986), including multiple stream matching. Besides, a new feature called Lifting Partitioning was introduced to the global optimizer RYSIA (Faria and Bagajewicz, 2011, 2012a, b) to help the lower bound of the model render higher values. Except the hyperstructure of Floudas and Ciric (1989), another significant and widely studied model was proposed by Yee and Grossmann (1990), based on the stage-wise superstructure. Due to the isothermal mixing assumption, a simpler HEN model with all linear constraints was obtained. Later, many modified models were proposed to address the limitations of the stage-wise superstructure (Yee and Grossmann, 1990). Björk and Westerlund (2002) and Huang et al. (2012) removed the isothermal mixing assumption to get better results. Hasan et al. (2010) modified the model to accommodate non-isothermal phase changes. Isafiade and Fraser (2008), Ponce-Ortega et al. (2010) and Na et al. (2015) expanded the model to cases with multiple utilities. Huang and Karimi (2013) proposed a multistage superstructure with cross flows to overcome the limitations of the two above described superstructures presented by Floudas et al. (1986) and Yee and Grossmann (1990). Although the proposed model brought some novel features, the complex structure led to a complex model. Later, an effective Outer approximation (OA) based algorithm (Huang and Karimi, 2014) was presented to solve the large-scale HENS problems. Recently, Faria et al. (2015) presented a bound contraction method to solve the stage-wise superstructure model of HEN. Short et al. (2016) proposed a novel methodology for HENS with detailed exchanger design, combining the stage-wise superstructure and a heuristic-based design approach. Pavão et al. (2016a, b; 2017) presented meta-heuristic methods to solve HENS problems, based on a modified stage-wise superstructure.

As is well known, the stage-wise superstructure of Yee and Grossmann (1990) is popular due to the fact that the nonlinear items only appear in the objective function, which makes the model robust. However, several alternative structures were neglected owing to the isothermal mixing assumption and the structure feature of the stage-wise superstructure, such as non-isothermal mixing, stream by-pass, and so on. As stated, many stage-wise superstructure based models were proposed with new features. Most of them were formulated as mixed integer nonlinear programming (MINLP) models, with non-convexity of nonlinear constraints. It is hard to get a feasible solution without good initial points. In spite of the NP-hard nature (Furman and Sahinidis, 2001) of HENS problems, it is necessary and realistic to develop an efficient and robust model for HENS.

In this paper, a new HEN model revised from the transshipment model of Papoulias and Grossmann (1983) is developed. The traditional transshipment model is capable of determining the minimum number of heat transfer matches and the pair of streams involved in each heat transfer match. However, it cannot provide the HEN configuration directly. Although the transportation/transshipment type of MILP model from Barbaro and Bagajewicz (2005) could give the cost-optimal solutions in one step, there was no specific mass flow pattern to describe the HEN configuration. By the proposed new model, the HEN configuration can be obtained by the mass flow pattern in one step. Besides, stream splitting, stream by-passing, isothermal and non-isothermal mixing, and recycling flows are all included in the model. Multiple utilities are available not only in the last stage to adjust the final temperature but also in an intermediate stage. Above all, in the case of allowing non-isothermal mixing, no nonlinear items appear in the constraints, while the objective function is nonlinear. The proposed model is formulated as a MINLP problem. Six examples are illustrated to demonstrate the applicability of the proposed model.

2. Problem statement

The general problem of HENS can be stated as follows. Given are a set of hot process streams (HP) to be cooled and a set of cold process streams (CP) to be heated. The flowrate, supply (starting) temperature, and target (ending) temperature of each hot process stream and each cold process stream are specified. A set of hot utilities (HU) and a set of cold utilities (CU) are given, as well as their corresponding temperatures. The objective is to design the HEN by minimizing the total annual cost. It is necessary to determine the number of heat exchangers, the heat load of each heat exchanger, the operating temperature of each heat exchanger, the flowrate of streams involved each heat exchanger, and the hot/cold utility consumption.

Let I represents hot streams (process streams and utilities), m represents cold streams (process streams and utilities). Define:

- $H=(I|I \text{ is a hot stream})$
- $HP=(I|I \text{ is a hot process stream})$
- $HU=(I|I \text{ is a hot utility stream})$

$H=HP \cup HU$

$C=(m|m \text{ is a cold stream})$

$CP=(m|m \text{ is a cold process stream})$

$CU=(m|m \text{ is a cold utility stream})$

$C=CP \cup CU$

3. Mathematical model

3.1. Partition of temperature intervals

The first step for the proposed method is to partition the entire temperature range of all streams into K temperature intervals. The specific steps can be found in Appendix A. An example with three hot streams and three cold streams is illustrated in Fig. 1. In this example, hot stream $l3$ and cold stream $m3$ are hot utility and cold utility respectively. Fig. 1-(a) shows that six temperatures (the circled ones) are chosen to construct the temperature intervals. Then, the starting temperature level (ns) and the ending temperature level (ne) for cold streams and hot streams are determined, as shown in Fig. 1-(b). For both cases with single utility and multiple utilities, it is assumed that there is one hot utility in the first temperature interval (temperature interval 1) to heat up cold process

streams and there is one cold utility in the last temperature interval (temperature interval K) to cool down hot process streams.

3.2. Mass flow pattern in HEN

After the location of each stream in the temperature intervals is determined, the transshipment model of Papoulias and Grossmann (1983) is extended to represent the HEN. The disadvantage of the traditional transshipment model is that it only indicates the heat transfer matches and the heat load of each heat transfer match rather than provide the HEN configuration, which should include the flowrate of hot/cold streams involved in each heat transfer match, the heat load of each heat transfer match, and the distribution of heat transfer matches. The reason is that only heat flow pattern is considered but not mass flow pattern in the traditional transshipment model. Heat flow pattern only describes how energy (heat) flows in the temperature intervals and how it flows from hot streams to cold streams, without how to distribute the streams in the HEN, such as splitting, mixing, and by-pass, namely mass flow pattern. As shown in Fig. 2-(a), each hot process stream l may exchange heat in every temperature interval below its starting temperature level ns_l , even the temperature intervals below its

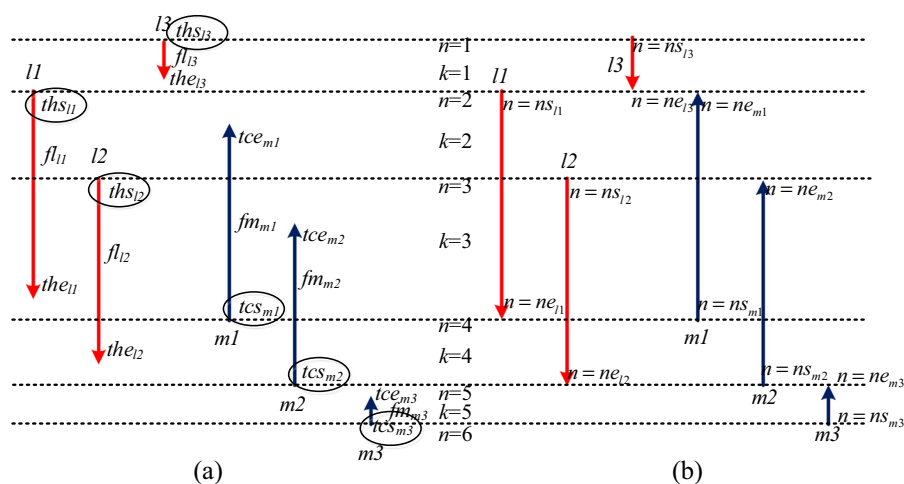


Fig. 1. Partition of temperature intervals.

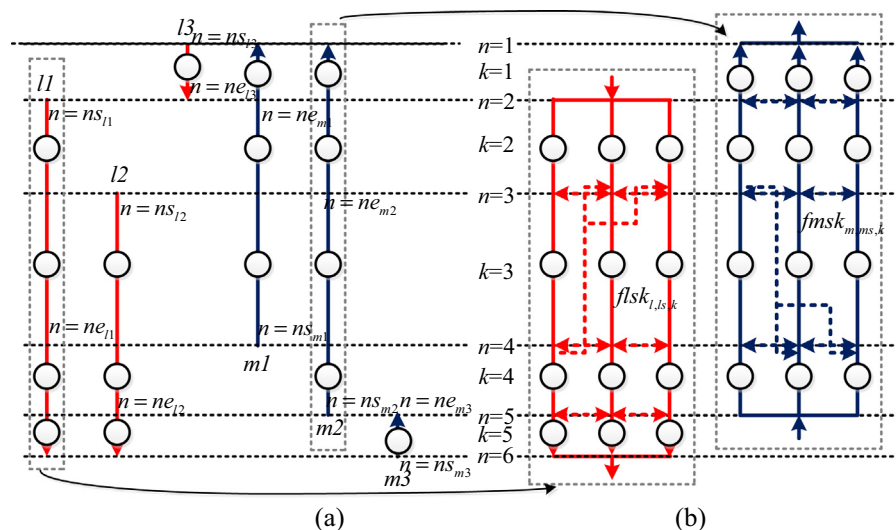


Fig. 2. Mass flow pattern and heat flow pattern in HEN.

ending temperature level ne_l . Since the temperature of all temperature levels are known, it is easy to keep the energy balance for each hot process stream, which will be introduced in the heat flow pattern. Similarly, each cold process stream m may exchange heat in every temperature interval above its starting temperature level ns_m , including the temperature intervals above its ending temperature level ne_m . It is assumed that each hot/cold utility is only available in one temperature interval. Therefore, heat transfer matches only exist in one temperature interval for hot utility streams and cold utility streams. The circles in Fig. 2-(a) indicate the feasible temperature intervals where the heat transfer matches can exist. However, each hot (cold) process stream l (m) may exchange heat with more than one cold (hot) process streams m (l) in one temperature interval. For example, heat flow pattern indicates that hot process stream l_1 can exchange heat with cold process streams m_1 and m_2 in temperature interval 3. Without mass flow pattern, the flowrate of streams involved in each heat transfer match cannot be determined. Besides, the distribution of heat transfer matches in one temperature interval can be series or parallel. The distribution among entire temperature intervals might be more complicated owing to the different distributions in each temperature interval. The HEN configuration is not specific without the mass flow pattern.

Therefore, in the proposed model, both heat flow pattern and mass flow pattern are considered so that the HEN can be obtained directly. Each hot (cold) stream l (m) should be split into several substreams ls (ms) in the mass flow pattern. Each hot (cold) process substream ls (ms) is assumed to exchange heat with at most one cold (hot) substream ms (ls) in each temperature interval. The mass flow pattern of a hot process stream and a cold process stream is illustrated in Fig. 2-(b), where each circle represents only one heat transfer match for a hot (cold) process substream ls (ms). Since there is only one heat transfer match for each process substream in one temperature interval, more substreams stand for more HEN configuration possibilities. The number of substreams can be determined according to the scale of the HENS problem. The minimum number must be one, while the maximum number of hot (cold) substreams equals the number of cold (hot) process streams. It is assumed that hot utility and cold utility are always utilized in parallel not in series. Since the inlet and outlet temperatures of hot utility and cold utility are constant, the only variables needed to be determined are the consumptions. Therefore, only one substream for each utility is enough. When there are more than one heat transfer matches for the utility substream, it means that the utility is consumed by more than one process streams. Besides, stream splitting, isothermal mixing, and non-isothermal mixing are all allowed between hot process substreams or cold process substreams, represented by the dash lines in Fig. 2-(b). The dashed lines with double side arrows mean that the mass flow may happen in either directions. The mass flow between all substreams may exist, although only two mass flows are present for each temperature interval in Fig. 2-(b). Except the mass flows between hot substreams in the same temperature levels, there are recycling mass flows from low temperature levels to high temperature levels for hot substreams. Similarly, there are recycling mass flows from high temperature levels to low temperature levels for cold substreams. For simplicity, Fig. 2-(b) shows only two recycling flows for hot substreams and cold substreams. Note that the recycling mass flow leads to direct heat transfer, which is not considered in the traditional transshipment model. Direct heat transfer means non-isothermal mixing that causes nonlinear terms in the model constraints. Yee and Grossmann (1990) avoided non-isothermal mixing to keep the constraints linear. The proposed model is able to include non-isothermal mixing without introducing nonlinear terms in the constraints, as shown in the following equations.

The mass balance equations for the mass flow pattern are given by Eqs. (1–12). The flowrate sum of substreams belonging to hot process stream l is equal to the flowrate of hot process stream l in its starting temperature interval ($k = ns_l$), given by Eq. (1). In the other temperature intervals ($k > ns_l$), The flowrate sum of substreams is equal to the flowrate sum of hot process stream l and the importing mass flows, given by Eq. (2). Besides, the mass balance between adjacent temperature intervals for hot substreams is given by Eq. (3). The flowrate of hot substream l, ls in temperature interval k must be equal to the flowrate sum of hot substream l, ls in temperature interval $k-1$ and the importing mass flows, subtracting the exporting mass flows. Similarly, the mass balance for cold substreams is ensured by Eqs. (4–6). The constraints for the flowrate of mass flows between substreams are given by Eqs. (7) and (8). The lower bound, the upper bound, and the specific values for the flowrate of process substreams and mass flows between substreams are given by Eqs. (9–12). Note that since the hot and cold utility consumptions are represented by the heat load not the flowrate, there are no mass balance equations for hot utilities and cold utilities.

$$\sum_{ls \in LS} flsk_{l,ls,k} = fl_l \quad \forall l \in HP, k \in TI, k = ns_l \quad (1)$$

$$\sum_{ls \in LS} flsk_{l,ls,k} = fl_l + \sum_{l' \in LS, n' > ns_l} flr_{l,ls',n',ls,n} \quad \forall l \in HP, k \in TI, ns_l < k = n \quad (2)$$

$$flsk_{l,ls,k} = flsk_{l,ls,k-1} + \sum_{l' \in LS, n' \geq n} flr_{l,ls',n',ls,n} - \sum_{l' \in LS, n' \leq n} flr_{l,ls,n,ls',n'} \quad (3)$$

$$\sum_{ms \in MS} fmsk_{m,ms,k} = fm_m \quad \forall m \in CP, k \in TI, k = ns_m - 1 \quad (4)$$

$$\sum_{ms \in MS} fmsk_{m,ms,k} = fm_m + \sum_{ms' \in MS, n' < ns_m} fmr_{m,ms',n',ms,n} \quad (5)$$

$$\forall m \in C, k \in TI, k < ns_m - 1, n = k + 1$$

$$fmsk_{m,ms,k} = fmsk_{m,ms,k+1} + \sum_{ms' \in MS, n' \leq n} fmr_{m,ms',n',ms,n} - \sum_{ms' \in MS, n' \geq n} fmr_{m,ms,n,ms',n'} \quad (6)$$

$$\sum_{l' \in LS, n' \geq n' > ns_l} flr_{l,ls,n,ls',n'} \leq flsk_{l,ls,k} + \sum_{l' \in LS, n' \leq n'} flr_{l,ls',n',ls,n} \quad (7)$$

$$\forall l \in HP, (l, ls) \in HLS, k \in TI, n \in TL, n > ns_l, k = n - 1$$

$$\sum_{ms' \in MS, ns_m > n' \geq n} fmr_{m,ms,n,ms',n'} \leq fmsk_{m,ms,k} + \sum_{ms' \in MS, n' \leq n} fmr_{m,ms',n',ms,n} \quad (8)$$

$$\forall m \in CP, (m, ms) \in CMS, k \in TI, n \in TL, n < ns_l, k = n$$

$$0 \leq flsk_{l,ls,k} \leq fl_l \quad \forall (l, ls) \in HLS, k \in TI$$

$$flsk_{l,ls,k} = \begin{cases} 0 & \forall (l, ls) \in HLS, l \in HP, k \in TI, k < ns_l \\ 0 & \forall (l, ls) \notin HLS, k \in TI \\ 0 & \forall (l, ls) \in HLS, l \in HU, k \in TI \end{cases} \quad (9)$$

$$0 \leq fmsk_{m,ms,k} \leq fm_m \quad \forall (m, ms) \in CMS, k \in TI$$

$$fmsk_{m,ms,k} = \begin{cases} 0 & \forall (m, ms) \in CMS, m \in CP, k \in TI, k \geq ns_m \\ 0 & \forall (m, ms) \notin CMS, k \in TI \\ 0 & \forall (m, ms) \in CMS, m \in CU, k \in TI \end{cases} \quad (10)$$

$$0 \leq flr_{l,ls',n',ls,n} \leq fl_l \quad \forall (l, ls), (l, ls') \in \text{HLS}, n, n' \in \text{TL}$$

$$flr_{l,ls',n',ls,n} = \begin{cases} 0 & \forall n \leq ns_l \text{ or } n' \leq ns_l \\ 0 & \forall n = N_k + 1 \text{ or } n' = N_k + 1 \\ 0 & \forall ls = ls' \\ 0 & \forall n > n' \\ 0 & \forall (l, ls) \notin \text{HLS or } (l, ls') \notin \text{HLS} \\ 0 & \forall l \in \text{HU} \end{cases} \quad (11)$$

$$0 \leq fmr_{m,ms',n',ms,n} \leq fm_m \quad \forall (m, ms), (m, ms') \in \text{CMS}, n, n' \in \text{TL}$$

$$fmr_{m,ms',n',ms,n} = \begin{cases} 0 & \forall n \geq ns_m \text{ or } n' \geq ns_m \\ 0 & \forall n = 1 \text{ or } n' = 1 \\ 0 & \forall ms = ms' \\ 0 & \forall n < n' \\ 0 & \forall (m, ms) \notin \text{CMS or } (m, ms') \notin \text{CMS} \\ 0 & \forall m \in \text{CU} \end{cases} \quad (12)$$

As shown in Fig. 2-(b), the flowrate of hot substreams and cold substreams may vary with the temperature intervals, owing to the mass flows between its own substreams, except hot utility substreams and cold utility substreams. Therefore, the binary variable $zlsk_{l,ls,n}$ ($zmsk_{m,ms,n}$) is proposed to represent the existence of importing mass flows to temperature level n of substream l,ls (m,ms). The binary variable $mlsk_{l,ls,n}$ ($mmsk_{m,ms,n}$) is proposed to show the existence of the exporting mass flows from temperature level n of substream l,ls (m,ms). When there is at least one importing mass flow from temperature level n' of hot substream l,ls' to temperature level n of hot substream l,ls existing, the binary variable $zlsk_{l,ls,n}$ will be set to 1 by Eq. (13). When there is at least one exporting mass flow from temperature level n of hot substream l,ls to temperature level n' of hot substream l,ls' existing, the binary variable $mlsk_{l,ls,n}$ will be set to 1 by Eq. (14). Otherwise, $mlsk_{l,ls,n}$ and $zlsk_{l,ls,n}$ will be set to 0, when minimizing the objective function. When considering cold substreams, the similar equations (Eqs. (15) and (16)) can be formulated. The specific values for $zlsk_{l,ls,n}$, $mlsk_{l,ls,n}$, $zmsk_{m,ms,n}$, and $mmsk_{m,ms,n}$ are given by Eqs. (17) and (18). The set of equations (Eqs. 13–18) is formulated to keep track of the flowrate change when stream splitting or mixing takes places. Besides, these four binary variables are essential for counting heat exchanger units, while $mlsk_{l,ls,n}$ and $mmsk_{m,ms,n}$ are also used to restrict stream splitting.

$$\sum_{ls' \in \text{LSn}' \geq n} flr_{l,ls',n',ls,n} \leq fl_l \times zlsk_{l,ls,n} \quad \forall l \in \text{HP}, (l, ls) \in \text{HLS}, n > ns_l \quad (13)$$

$$\sum_{ls' \in \text{LSn}' \geq n'} flr_{l,ls,n,ls',n'} \leq fl_l \times mlsk_{l,ls,n} \quad \forall l \in \text{HP}, (l, ls) \in \text{HLS}, n > ns_l \quad (14)$$

$$\sum_{ms' \in \text{MSn}' \geq n'} fmr_{m,ms',n',ms,n} \leq fm_m \times zmsk_{m,ms,n} \quad \forall m \in \text{CP}, (m, ms) \in \text{CMS}, n < ns_m \quad (15)$$

$$\sum_{ms' \in \text{MSn}' \geq n} fmr_{m,ms,n,ms',n'} \leq fm_m \times mmsk_{m,ms,n} \quad \forall m \in \text{CP}, (m, ms) \in \text{CMS}, n < ns_m \quad (16)$$

$$zlsk_{l,ls,n}, mlsk_{l,ls,n} = \begin{cases} 0 & \forall (l, ls) \in \text{HLS}, l \in \text{HP}, n \leq ns_l \text{ or } n = N \\ 0 & \forall (l, ls) \notin \text{HLS}, l \in \text{HP} \\ 0 & \forall l \in \text{HU} \end{cases} \quad (17)$$

$$zmsk_{m,ms,n}, mmsk_{m,ms,n} = \begin{cases} 0 & \forall (m, ms) \in \text{CMS}, m \in \text{CP}, n \geq ns_m \text{ or } n = 1 \\ 0 & \forall (m, ms) \notin \text{CMS}, m \in \text{CP} \\ 0 & \forall m \in \text{CU} \end{cases} \quad (18)$$

3.3. Heat flow pattern in HEN

In the heat flow pattern, each substream is considered as an individual stream. Fig. 3 shows the heat flow pattern for a hot substream and a cold substream in one temperature interval. The heat flow pattern for hot process substreams can be described as follows: the residual energy $rqhs_{l,ls,n}$ which cannot be utilized in the previous temperature interval (interval $k-1$) flows into the present temperature interval (interval k , $k = n$); the importing mass flow $flr_{l,ls',n',ls,n}$ flows from temperature level n' of hot substream l,ls' to temperature level n of hot substream l,ls ; the inherent energy $flsk_{l,ls,k}(tn_n - tn_{n+1})c_p$ represents supplied heat by hot substream l,ls in the present temperature interval; the heat load $q_{l,ls,m,ms,k}$ flows out to a cold substream m,ms by a heat transfer match with a cold process substream or a cold utility substream; the residual energy $rqhs_{l,ls,n+1}$ which cannot be utilized in the present temperature interval flows out to the next temperature interval (interval $k+1$). In temperature level n , the importing mass flows are mixed with the residual energy $rqhs_{l,ls,n}$. The residual energy $rqhs'_{l,ls,n}$ after mixing is given by Eq. (19). Since the importing mass flows can be from temperature levels lower than temperature level n , the residual energy $rqhs'_{l,ls,n}$ is defined as a positive variable, which can make sure it is still positive after mixing. Then, the energy balance in each temperature interval is given by Eq. (20). As is well known that, hot streams supply heat in the HEN while cold streams demand heat. In the proposed heat flow pattern, the residual energy of cold substreams is fictitious energy (positive), which actually represents the unsatisfied demand. The inherent energy $fmsk_{m,ms,k}(tn_n - tn_{n+1})c_p$ represents the heat demand of cold substream m,ms in the present temperature interval. The heat flow pattern for cold process substreams is similar with the one for hot process substreams, but in the opposite direction, given by Eqs. (21) and (22).

Energy balance for hot process substreams in each temperature interval:

$$rqhs'_{l,ls,n} = rqhs_{l,ls,n} - \sum_{ls' \in \text{LSn}' > n} flr_{l,ls',n',ls,n} \times (tn_n - tn_{n'}) \times c_p \quad \forall (l, ls) \in \text{HLS}, l \in \text{HP}, n \in \text{TL}, n \geq ns_l \quad (19)$$

$$flsk_{l,ls,k} \times (tn_n - tn_{n+1}) \times c_p + rqhs'_{l,ls,n} = rqhs_{l,ls,n+1} + \sum_{(m,ms) \in \text{CMS}, m \in \text{C}_k} q_{l,ls,m,ms,k} \quad \forall (l, ls) \in \text{HLS}, l \in \text{HP}, n \in \text{TL}, k \in \text{TI}, n = k, ns_l \leq k \quad (20)$$

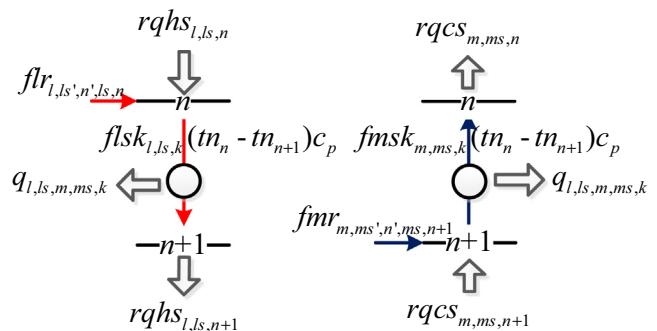


Fig. 3. Heat flow pattern in temperature interval k .

Energy balance for cold process substreams in each temperature interval:

$$rqcs'_{m,ms,n} = rqcs_{m,ms,n} - \sum_{ms' \in MSn' < n} \sum_{n' \in TL, n' < n} (fmr_{m,ms',n',ms,n} \times (tn_{n'} - tn_n) \times c_p) \quad \forall (m, ms) \in CMS, m \in CP, n \in TL, n \leq ns_m \quad (21)$$

$$\begin{aligned} fmsk_{m,ms,k} \times (tn_n - tn_{n+1}) \times c_p + rqcs'_{m,ms,n+1} \\ = rqcs_{m,ms,n} + \sum_{(l,ls) \in HLS, l \in H_k} q_{l,ls,m,ms,k} \quad \forall (m, ms) \in CMS, m \\ \in CP, n \in TL, k \in TI, n = k, k < ns_m \end{aligned} \quad (22)$$

In the proposed model, not only isothermal mixing but also non-isothermal mixing between hot process substreams or cold process substreams can happen, as well as stream splitting of hot or cold process substreams. Stream mixing of hot process substreams and cold process substreams is restrained by Eqs. (19) and (21) respectively, which means that it is feasible once $rqhs'_{l,ls,n}/rqcs'_{m,ms,n}$ is positive. Besides, stream splitting is forbidden unless $rqhs'_{l,ls,n}/rqcs'_{m,ms,n}$ is equal to 0, which makes sure the temperature of the splitting flow is equal to tn_n . When stream splitting happens in temperature level n of hot process substream l,ls ($mlsk_{l,ls,n} = 1$, given by Eq. (13)), $rqhs'_{l,ls,n}$ must be 0 by the constraint of Eq. (23). Similarly, $rqcs'_{m,ms,n}$ will be set to 0 by Eq. (24), when stream splitting happens in temperature level n of cold process substream m,ms . Note that, parameters $MRL_{l,n}$ and $MRM_{m,n}$ represent the maximum residual energy of hot process stream l and cold process stream m in the temperature level n respectively, given by Eqs. (25) and (26).

$$rqhs'_{l,ls,n} \leq MRL_{l,n} \times (1 - mlsk_{l,ls,n}) \quad \forall (l, ls) \in HLS, l \in HP, n \in TL, n > ns_l \quad (23)$$

$$rqcs'_{m,ms,n} \leq MRM_{m,n} \times (1 - mmsk_{m,ms,n}) \quad \forall (m, ms) \in CMS, m \in CP, n \in TL, n < ns_m \quad (24)$$

$$MRL_{l,n} = \begin{cases} fl_l \times c_p \times (ths_l - tn_n) & \forall l \in HP, n \in TL, n > ns_l, n \neq N \\ fl_l \times c_p \times (the_l - tn_n) & \forall l \in HP, n \in TL, n = N \end{cases} \quad (25)$$

$$MRM_{m,n} = \begin{cases} fm_m \times c_p \times (tn_n - tcs_m) & \forall m \in CP, n \in TL, n < ns_m, n \neq 1 \\ fm_m \times c_p \times (tn_n - tce_m) & \forall m \in CP, n \in TL, n = 1 \end{cases} \quad (26)$$

According to the rule mentioned in Appendix A, the temperature of the starting and ending temperature levels of hot streams and cold streams may not be exactly equal to their starting and ending temperatures. As shown in Fig. 4, one example with a hot stream and a cold stream is illustrated. Neither the starting temperature nor the ending temperature of these two streams is chose to construct the temperature intervals. The starting temperature level and the ending temperature level are determined according to the rule in Appendix A. The lines without heat transfer matches (circles) represent the location where the stream should be according to the starting and ending temperatures, while the lines with heat transfer matches represent the actual location in the heat flow pattern. Hence, in the starting temperature level, the residual energy of hot and cold process substreams is not zero, their values are given by Eqs. (27) and (28) respectively. Even when the temperature of starting temperature level is equal to the starting temperature, the residual energy of zero also can be obtained by Eq. (27) or Eq. (28) ($ths_l = tn_n$ or $tcs_m = tn_n$). The similar equations can be obtained

for the ending temperature. Instead of restricting the value of the residual energy of each substream, Eqs. (29) and (30) restrict the sum of the residual energy of all hot process substreams and all cold process substreams respectively. Under such circumstances, the ending temperature of substreams belonging to a same process stream can be different with the ending temperature of the process stream. Therefore, the mixing of substreams in the ending temperature level can be isothermal or non-isothermal, which will enrich the structure possibilities. The lower bound, the upper bound, and the specific values for $rqhs_{l,ls,n}$, $rqhs'_{l,ls,n}$, $rqcs_{m,ms,n}$, and $rqcs'_{m,ms,n}$ are given by Eqs. (31) and (32).

$$\begin{aligned} rqhs_{l,ls,n} = \frac{flsk_{l,ls,n}}{fl_l} \times fl_l \times c_p \times (ths_l - tn_n) \quad \forall (l, ls) \in HLS, l \\ \in HP, k \in TI, n \in TL, n = k = ns_l \end{aligned} \quad (27)$$

$$\begin{aligned} rqcs_{m,ms,n} = \frac{fmsk_{m,ms,n}}{fm_m} \times fm_m \times c_p \times (tn_n - tcs_m) \quad \forall (m, ms) \\ \in CMS, m \in CP, k \in TI, n \in TL, n = ns_l, k = n - 1 \end{aligned} \quad (28)$$

$$\sum_{ls \in LS} rqhs_{l,ls,n} = fl_l \times c_p \times (the_l - tn_n) \quad \forall l \in HP, n \in TL, n = N \quad (29)$$

$$\sum_{ms \in MS} rqcs_{m,ms,n} = fm_m \times c_p \times (tn_n - tce_m) \quad \forall m \in CP, n \in TL, n = 1 \quad (30)$$

$$\begin{aligned} 0 \leq rqhs_{l,ls,n} \leq MRL_{l,n} \quad \forall (l, ls) \in HLS, n \in TL \\ 0 \leq rqhs'_{l,ls,n} \leq MRL_{l,n} \quad \forall (l, ls) \in HLS, n \in TL \\ rqhs_{l,ls,n}, rqhs'_{l,ls,n} = \begin{cases} 0 & \forall (l, ls) \in HLS, l \in HP, n \in TL, n < ns_l \\ 0 & \forall (l, ls) \notin HLS, n \in TL \\ 0 & \forall l \in HU, n \in TL \end{cases} \end{aligned} \quad (31)$$

$$\begin{aligned} 0 \leq rqcs_{m,ms,n} \leq MRM_{m,n} \quad \forall (m, ms) \in CMS, n \in TL \\ 0 \leq rqcs'_{m,ms,n} \leq MRM_{m,n} \quad \forall (m, ms) \in CMS, n \in TL \\ rqcs_{m,ms,n}, rqcs'_{m,ms,n} = \begin{cases} 0 & \forall (m, ms) \in CMS, m \in CP, n \in TL, n > ns_m \\ 0 & \forall (m, ms) \notin CMS, n \in TL \\ 0 & \forall m \in CU, n \in TL \end{cases} \end{aligned} \quad (32)$$

Besides, the utility consumptions can be determined with the sum of the heat load of the heat transfer matches between process substreams and utility substreams. In the proposed model, only the heat load but not the flowrate is adopted to quantify the hot and cold utility consumptions.

Hot utility l and cold utility m consumptions are given by:

$$hu_l = \sum_{ls \in LS(m,ms) \in CMS, m \in CP, k \in TI, ns_l \leq k < ne_l, k < ns_m} q_{l,ls,m,ms,k} \quad \forall l \in HU \quad (33)$$

$$cu_m = \sum_{ms \in MS(l,ls) \in HLS, l \in HP, k \in TI, ns_l \leq k < ne_l, k < ns_m} q_{l,ls,m,ms,k} \quad \forall m \in CU \quad (34)$$

Logical constraints for heat loads are given by Eq. (35), where the binary variables $z_{l,ls,m,ms,k}$ represent the existence of the heat transfer matches between hot substream l,ls and cold substream m,ms in temperature interval k . The parameter $\Omega_{l,m,k}$ represents the upper bound for heat transfer load of the heat transfer matches between hot stream l and cold stream m in temperature interval k , given by Eq. (36).

$$\begin{aligned} q_{l,ls,m,ms,k} - \Omega_{l,m,k} \times z_{l,ls,m,ms,k} \leq 0 \\ \forall (l, ls) \in HLS, l \in H, (m, ms) \in CMS, m \in C, k \in TI, ns_l \leq k < ns_m \end{aligned} \quad (35)$$

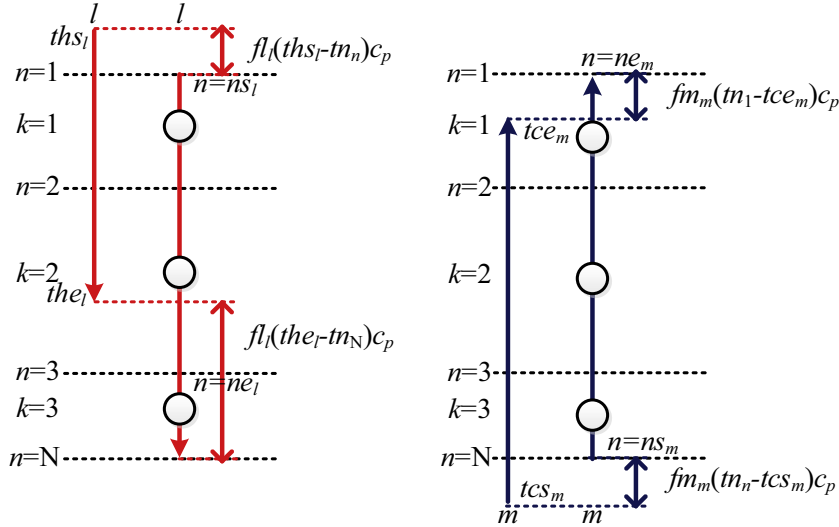


Fig. 4. Residual energy of streams in the starting and ending temperature levels.

$$\Omega_{l,m,k} = \begin{cases} \min(HLK_{l,k}, CLK_{m,k}, HL_l, CL_m) & \forall l \in HP, m \in CP \\ \min(HLK_{l,k}, HL_l) & \forall l \in HP, m \in CU \\ \min(CLK_{m,k}, CL_m) & \forall l \in HU, m \in CP \\ 0 & \forall l \in HU, m \in CU \end{cases}$$

$$HL_l = fl_l \times c_p \times (ths_l - the_l) \quad \forall l \in HP$$

$$HLK_{l,k} = \begin{cases} 0 & \forall l \in HP, k \in TI, ns_l > k \\ fl_l \times c_p \times (ths_l - tn_n) & \forall l \in HP, k \in TI, ns_l \leq k, n = k + 1 \end{cases}$$

$$CL_m = fm_m \times c_p \times (tce_m - tcs_m) \quad \forall m \in CP$$

$$CLK_{m,k} = \begin{cases} 0 & \forall m \in CP, k \in TI, ns_m \leq k \\ fm_m \times c_p \times (tn_n - tcs_m) & \forall m \in CP, k \in TI, ns_m > k, n = k \end{cases} \quad (36)$$

Note that, there is at most one heat transfer match in one temperature interval for each hot/cold process substream, given by Eqs. (37) and (38) respectively. For each hot/cold process stream, there is at least one heat transfer match for its substreams in the whole temperature intervals, given by Eqs. (39) and (40) respectively. Besides, it is assumed that hot process streams are only cooled down by either cold process streams or cold utility M (M represents the cold utility with the smallest starting temperature) in the last temperature interval K, given by Eq. (41). Similarly, it is assumed that cold process streams are only heated up by either hot process streams or hot utility L (L represents the hot utility with the largest starting temperature) in the first temperature interval 1, given by Eq. (42).

$$\sum_{(m,ms) \in CMS} z_{l,ls,m,ms,k} \leq 1 \quad \forall (l,ls) \in HLS, l \in HP, k \in TI, ns_l \leq k \quad (37)$$

$$\sum_{(l,ls) \in HLS} z_{l,ls,m,ms,k} \leq 1 \quad \forall (m,ms) \in CMS, m \in CP, k \in TI, k < ns_m \quad (38)$$

$$\sum_{ls \in LS} \sum_{(m,ms) \in CMS} z_{l,ls,m,ms,k} \geq 1 \quad \forall l \in HP \quad (39)$$

$$\sum_{ms \in MS} \sum_{(l,ls) \in HLS} z_{l,ls,m,ms,k} \geq 1 \quad \forall m \in CP \quad (40)$$

$$z_{l,ls,M,ms,K} + z_{l',ls',M',ms',K} \leq 1 \quad \forall (l,ls) \in HLS, l \in HP, (m',ms') \in CMS, m' \in CP \quad (41)$$

$$z_{L,ls,m,ms,1} + z_{l',ls',m,ms,1} \leq 1 \quad \forall (l',ls') \in HLS, l' \in HP, (m,ms) \in CMS, m \in CP \quad (42)$$

3.4. Temperature difference for heat transfer

In the heat flow pattern, there are three kinds of heat transfer matches, as shown in Fig. 5-1, 2, and 3, which are heat transfer matches between a hot process substream and a cold process substream, a hot process substream and a cold utility substream, and a hot utility substream and a cold process substream, respectively. Although the temperature of each temperature level is constant, the temperature of each substream in each temperature level is not always equal to the fixed value, since there may be residual energy existing. When a hot process substream flows into temperature interval k ($k=n$), with non-zero residual energy $rqhs'_{l,ls,n}$, the inlet temperature $til_{l,ls,k}$ is not equal to but larger than the temperature of temperature level n (tn_n). Besides, when a hot process substream flows out from temperature interval k ($k=n$), with non-zero residual energy $rqhs_{l,ls,n+1}$, the outlet temperature $tol_{l,ls,k}$ is larger than the temperature of temperature level $n+1$ (tn_{n+1}). For hot utility substreams, the inlet temperature $til_{l,ls,k}$ and the outlet temperature $tol_{l,ls,k}$ are equal to the starting temperature and the ending temperature of hot utility l , since every utility is assumed to be available in only one temperature interval. Thus, the inlet temperature $til_{l,ls,k}$ and the outlet temperature $tol_{l,ls,k}$ of hot substreams are given by Eqs. (43) and (44) respectively. Similarly, Eqs. (45) and (46) are formulated for cold substreams. With the inlet and outlet temperatures of hot substreams and cold substreams involved in each heat transfer match, the logarithmic mean temperature difference (LMTD) $dt_{l,ls,m,ms,k}$ can be obtained by Chen's LMTD approximation (Chen, 1987), given by Eq. (47). As shown in Fig. 5-4, when hot process streams exchange heat with cold utility streams in the last temperature interval K, all heat transfer matches can be treated as one heat transfer match. As a result, all hot substreams are mixed before being cooled down by the cold utility stream so that at most one cooler is needed for each hot process stream in the last temperature interval K. As for cold process streams, substreams are mixed before being heated up by the hot utility stream in the first temperature interval 1, as shown in Fig. 5-5. The temperature differences are given by Eqs. (48) and (49) respectively. In order to keep the linearity of constraints, these seven nonlinear equations (Eqs. 43–49) are all substituted into the objective function.

$$til_{l,ls,k} = \begin{cases} tn_n + \Delta T_{\min} + \frac{rqhs'_{l,ls,n}}{fl_{ls,k} \times c_p} & \forall (l,ls) \in HLS, l \in HP, k \in TI \\ ths_l & \forall (l,ls) \in HLS, l \in HU, k \in TI \end{cases} \quad (43)$$

$$tol_{l,ls,k} = \begin{cases} tn_{n+1} + \Delta T_{\min} + \frac{rqhs_{l,ls,n+1}}{f_{lsk_{l,ls,k} \times c_p}} & \forall (l, ls) \in \text{HLS}, l \in \text{HP}, k \in \text{TI} \\ the_l & \forall (l, ls) \in \text{HLS}, l \in \text{HU}, k \in \text{TI} \end{cases} \quad (44)$$

$$tom_{m,ms,k} = \begin{cases} tn_n - \frac{rqhs_{m,ms,n}}{f_{msk_{m,ms,k} \times c_p}} & \forall (m, ms) \in \text{CMS}, m \in \text{CP}, k \in \text{TI} \\ tce_m & \forall (m, ms) \in \text{CMS}, m \in \text{CU}, k \in \text{TI} \end{cases} \quad (45)$$

$$tim_{m,ms,k} = \begin{cases} tn_{n+1} - \frac{rqhs'_{m,ms,n+1}}{f_{msk_{m,ms,k} \times c_p}} & \forall (m, ms) \in \text{CMS}, m \in \text{CP}, k \in \text{TI} \\ tcs_m & \forall (m, ms) \in \text{CMS}, m \in \text{CU}, k \in \text{TI} \end{cases} \quad (46)$$

$$dt_{l,ls,m,ms,k} = \left[\frac{(til_{l,ls,k} - tom_{m,ms,k})(tol_{l,ls,k} - tim_{m,ms,k})}{\times 0.5 \times (til_{l,ls,k} - tom_{m,ms,k} + tol_{l,ls,k} - tim_{m,ms,k})} \right]^{\frac{1}{3}} \quad (47)$$

$$\forall (l, ls) \in \text{HLS}, l \in \text{H}, l \neq L, (m, ms) \in \text{CMS}, m \in \text{C}, m \neq M, k \in \text{TI}$$

$$dthu_m = \left[\frac{(ths_L + \Delta T_{\min} - tce_m) \left(the_L + \Delta T_{\min} - tn_2 + \frac{\sum ms}{f_{m_m} \times c_p} \right)}{\times 0.5 \times \left(ths_L + \Delta T_{\min} - tce_m + the_L + \Delta T_{\min} - tn_2 + \frac{\sum ms}{f_{m_m} \times c_p} \right)} \right]^{\frac{1}{3}} \quad \forall m \in \text{CP} \quad (48)$$

$$dtcu_l = \left[\frac{\left(\frac{\sum ls}{f_{l_l} \times c_p} + \Delta T_{\min} + tn_{N-1} - tce_M \right) (\Delta T_{\min} + the_l - tcs_M)}{\times 0.5 \times \left(\frac{\sum ls}{f_{l_l} \times c_p} + \Delta T_{\min} + tn_{N-1} - tce_M + \Delta T_{\min} + the_l - tcs_M \right)} \right]^{\frac{1}{3}} \quad \forall l \in \text{HP} \quad (49)$$

3.5. Identification of heat exchangers

As is well known, the traditional transshipment model can indicate the number of heat transfer matches and the pair of streams involved in each heat transfer match. However, the sum of heat transfer matches might not be identical to the sum of heat exchangers. Two heat transfer matches between one hot substream and one cold substream can be merged into one heat exchanger, when they are in two adjacent temperature intervals and the flowrate of two substreams remain unchanged in these two temperature intervals. In this study, Eqs. 50–52 are proposed to count the heat exchanger number exactly, where the binary variable $zz_{l,ls,m,ms,k}$ represents the existence of the heat exchanger between hot substream l,ls and cold substream m,ms in temperature interval k . There may be several heat transfer matches between one hot substream and one cold substream, let's count heat exchangers from the starting temperature interval of the hot substream. For the starting temperature interval ($k = ns_i$) of hot stream l , $zz_{l,ls,m,ms,k}$ is equal to $z_{l,ls,m,ms,k}$, given by Eq. (50). It means that a heat exchanger exists once a heat transfer match exists in the starting temperature interval. For the other temperature intervals, the value of $zz_{l,ls,m,ms,k}$ depends on two conditions: first, the

existence of the heat transfer match in temperature interval k (the value of $z_{l,ls,m,ms,k}$); second, the feasibility of being merged into temperature interval $k-1$. It is obvious that the first condition is a prerequisite for the second one. Once no heat transfer match exists in temperature interval k , there is no need to determine whether the heat transfer match can be merged into temperature interval $k-1$. The first condition is determined by Eq. (51)-(a), while the second condition is determined by Eqs. (51)-(b), (c), (d), (e), and (f). To illustrate how the set of constraints works, six cases are presented in Fig. 6 and Table 1. The dashed line with double side arrows in Fig. 6 means that the mass flow may happen in either directions. The flowrates of hot substream l,ls and cold substream m,ms are shown in the first row and the fourth row of Table 1 respectively, followed by the binary variables denoting the consistency of the flowrates, which can be obtained by Eqs. (13–16). Note that the flowrates in Table 1 are hypothetical for the illustration. The right-side value of the inequalities (Eq. (51)) are shown in the following rows, while the binary variable $zz_{l,ls,m,ms,k}$ is in the last row. The value of $zz_{l,ls,m,ms,k}$ is obtained according to the right-side value of six inequalities, while the contributing ones are marked bold.

Case 1: when there is no heat transfer match in temperature interval k ($z_{l,ls,m,ms,k} = 0$), as shown in Fig. 6-(1), it is so obvious that there is no heat exchanger in this temperature interval. The right-side value of Eq. (51)-(a) is equal to 0. Therefore, the binary variable $zz_{l,ls,m,ms,k}$ is always forced to 0, by Eq. (51)-(a). The other constraints become trivial, as shown in Table 1.

Case 2: when one heat transfer match exists in temperature interval k but none in temperature interval $k-1$, as shown in Fig. 6-(2). Since there is no heat transfer match in temperature interval $k-1$, the heat transfer match in temperature interval k cannot be merged with the previous one. The right-side value of Eq. (51)-(b) is equal to 1. Thus, the binary variable $zz_{l,ls,m,ms,k}$ is always forced to 1 by Eq. (51)-(b). There will be one heat exchanger in temperature interval k . Once Eq. (51)-(b) is activated, the constraints for the flowrate change between two temperature intervals for the hot substream and the cold substream (Eqs. (51)-(c), (d), (e), and (f)) become trivial. No matter what the value of $z_{lsk_l,ls,k}$, $z_{msk_m,ms,k}$, $m_{lsk_l,ls,k}$, and $m_{msk_m,ms,k}$ will be, the value of $zz_{l,ls,m,ms,k}$ will not be changed, as shown in Table 1.

Case 3, 4, 5 and 6, when there are two heat transfer matches in two adjacent temperature intervals, as shown in Fig. 6-(3), (4), (5) and (6), the value of $zz_{l,ls,m,ms,k}$ is determined by Eqs. (51)-(c), (d),

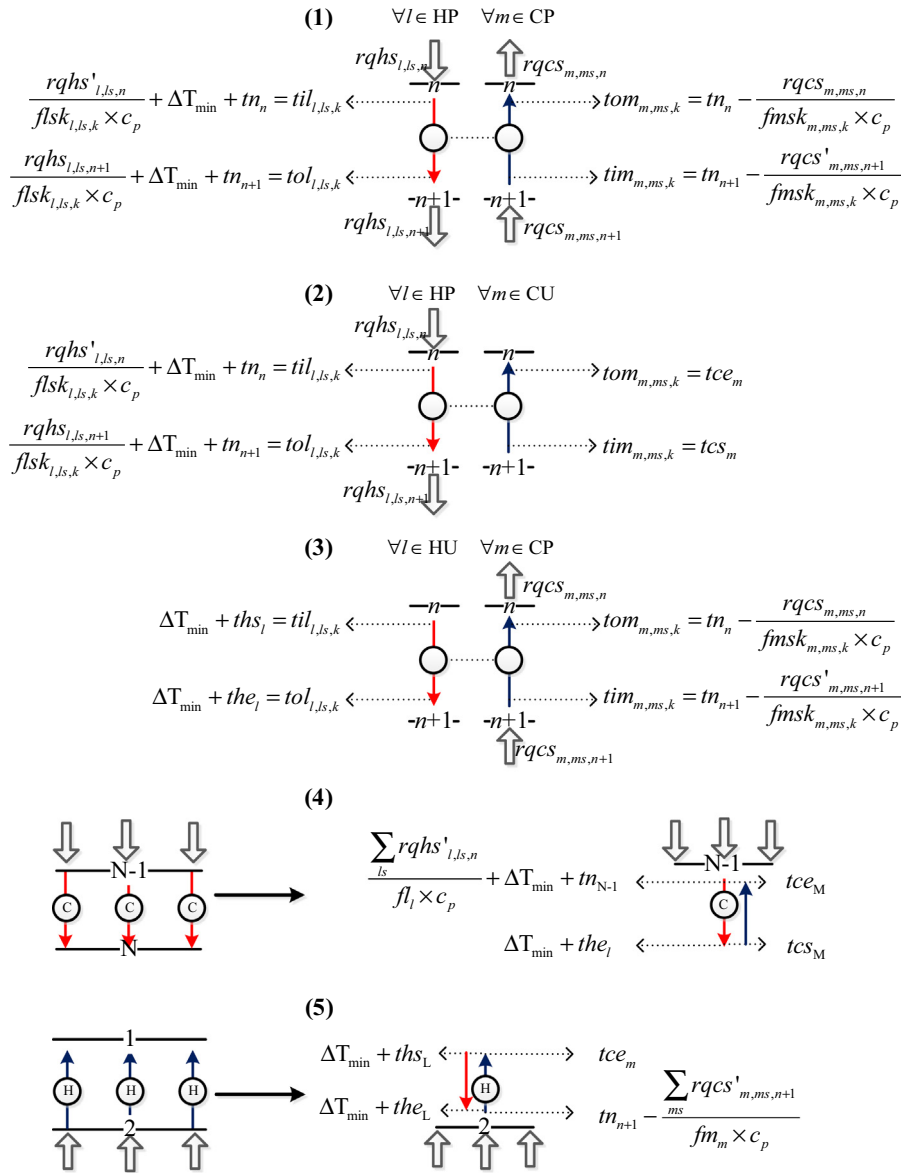


Fig. 5. Illustration for temperature differences.

(e) and (f). As mentioned before, only when the flowrate of the hot substream and the cold substream remain unchanged in two adjacent temperature intervals, two heat transfer matches can be merged together. Otherwise, one more heat exchanger will be needed. In case 3, the flowrate of hot and cold substreams both remain unchanged in these two temperature intervals, $zlsk_{l,ls,k}$, $zmsk_{m,ms,k}$, $mlsk_{l,ls,k}$, and $mmsk_{m,ms,k}$ are all equal to 0. Thus, the right-side value of the inequalities (Eqs. (51)–(c), (d), (e) and (f)) will be 0. Then, the value of $zz_{l,ls,m,ms,k}$ is set to 0 when minimizing the objective function. In cases 4, 5 and 6, when there is at least one exporting mass flow from hot or cold substreams or at least one importing mass flow to hot or cold substreams existing, at least one of these four binary variables $zlsk_{l,ls,k}$, $zmsk_{m,ms,k}$, $mlsk_{l,ls,k}$, and $mmsk_{m,ms,k}$ will be 1. Table 1 shows that the binary variable $zz_{l,ls,m,ms,k}$ will be forced to 1 by one of Eqs. (51)–(c), (d), (e), and (f). Thus, there is one more heat exchanger needed.

The lower bound, the upper bound, and the specific values for $zz_{l,ls,m,ms,k}$, $z_{l,ls,m,ms,k}$, and $q_{l,ls,m,ms,k}$ are given by Eq. (52).

$$zz_{l,ls,m,ms,k} = z_{l,ls,m,ms,k} \quad \forall (l, ls) \in \text{HLS}, (m, ms) \in \text{CMS}, k \in \text{TI}, k = ns_l \quad (50)$$

$$\left. \begin{aligned} (a) \quad & zz_{l,ls,m,ms,k} \leq z_{l,ls,m,ms,k} \\ (b) \quad & zz_{l,ls,m,ms,k} \geq z_{l,ls,m,ms,k} - z_{l,ls,m,ms,k-1} \\ (c) \quad & zz_{l,ls,m,ms,k} \geq zlsk_{l,ls,n} + z_{l,ls,m,ms,k} - 1 \\ (d) \quad & zz_{l,ls,m,ms,k} \geq mlsk_{l,ls,n} + z_{l,ls,m,ms,k} - 1 \\ (e) \quad & zz_{l,ls,m,ms,k} \geq zmsk_{m,ms,n} + z_{l,ls,m,ms,k} - 1 \\ (f) \quad & zz_{l,ls,m,ms,k} \geq mmsk_{m,ms,n} + z_{l,ls,m,ms,k} - 1 \end{aligned} \right\} \quad (51)$$

$$\forall (l, ls) \in \text{HLS}, (m, ms) \in \text{CMS}, k \in \text{TI}, n \in \text{TI}, ns_l < k = n$$

$$0 \leq q_{l,ls,m,ms,k} \leq \Omega_{l,m,k} \quad \forall (l, ls) \in \text{HLS}, (m, ms) \in \text{CMS}, k \in \text{TI}$$

$$\left. \begin{aligned} & zz_{l,ls,m,ms,k} \\ & z_{l,ls,m,ms,k} \\ & q_{l,ls,m,ms,k} \end{aligned} \right\} = \begin{cases} 0 & \forall (l, ls) \in \text{HLS}, l \in \text{HP}, (m, ms) \in \text{CMS}, ms \in \text{CP}, \\ & k \in \text{TI}, k < ns_l \text{ or } k \geq ns_m \\ 0 & \forall (l, ls) \notin \text{HLS} \text{ or } (m, ms) \notin \text{CMS}, k \in \text{TI} \\ 0 & \forall l \in \text{HU}, m \in \text{CU}, k \in \text{TI} \\ 0 & \forall (l, ls) \in \text{HLS}, l \in \text{HU}, k \in \text{TI}, k < ns_l \text{ or } k \geq ne_l \\ 0 & \forall (m, ms) \in \text{CMS}, m \in \text{CU}, k \in \text{TI}, k < ne_m \text{ or } k \geq ns_m \end{cases} \quad (52)$$

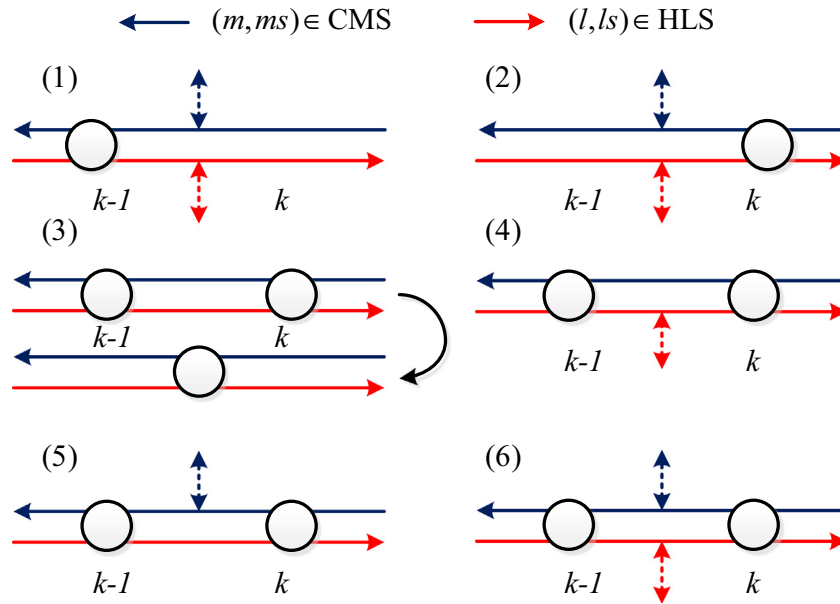


Fig. 6. Illustration for heat exchanger counting.

Table 1
Illustration for heat exchanger counting (data).

Case	1		2		3		4/5/6	
	k-1	k	k-1	k	k-1	k	k-1	k
$flsk_{l,ls,k}$	50	45	50	55	50	50	50	45/50/55
$zlsk_{l,ls,k}$	/	0	/	1	/	0	/	0/0/1
$mlsk_{l,ls,k}$	/	1	/	0	/	0	/	1/0/0
$fmsk_{m,ms,k}$	50	55	50	45	50	50	50	50/45/45
$zmsk_{m,ms,k}$	/	1	/	0	/	0	/	0/0/0
$mmsk_{m,ms,k}$	/	0	/	1	/	0	/	0/1/1
(a) $Z_{l,ls,m,ms,k}$	1	≤ 0	0	≤ 1	1	≤ 1	1	≤ 1
(b) $Z_{l,ls,m,ms,k} - Z_{l,ls,m,ms,k-1}$	/	≥ -1	/	≥ 1	/	≥ 0	/	≥ 0
(c) $zlsk_{l,ls,k} + Z_{l,ls,m,ms,k-1}$	/	≥ -1	/	≥ 1	/	≥ 0	/	$\geq 0/0/1$
(d) $mlsk_{l,ls,k} + Z_{l,ls,m,ms,k-1}$	/	≥ 0	/	≥ 0	/	≥ 0	/	$\geq 1/0/0$
(e) $zmsk_{m,ms,k} + Z_{l,ls,m,ms,k-1}$	/	≥ 0	/	≥ 0	/	≥ 0	/	$\geq 0/0/1$
(f) $mmsk_{m,ms,k} + Z_{l,ls,m,ms,k-1}$	/	≥ -1	/	≥ 1	/	≥ 0	/	$\geq 0/1/0$
$ZZ_{l,ls,m,ms,k}$	/	0	/	1	/	0	/	1/1/1

As mentioned above, all substreams of hot process streams are mixed before being cooled down by cold utility M in the last temperature interval K. Therefore, no matter how many heat transfer matches exist between hot process stream l and cold utility M, there is at most one cooler for each hot process stream in the last temperature interval K, given by Eq. (53). The binary variable zcu_l represents the existence of cold utility M for hot process stream l in the last temperature interval K. Similarly, there is at most one heater for each cold process stream in the first temperature interval 1, given by Eq. (54), where the binary variable zhu_m represents the existence of hot utility L for cold process stream m in the first temperature interval 1.

$$zcu_l \geq ZZ_{l,ls,M,ms,K} \quad \forall (l, ls) \in \text{HLS}, l \in \text{HP} \quad (53)$$

$$zhu_m \geq ZZ_{L,l,ls,m,1} \quad \forall (m, ms) \in \text{CMS}, m \in \text{CP} \quad (54)$$

After counting the number of heat exchangers, the heat load of each heat exchanger should be calculated. There might be more than one heat exchangers between a hot process substream and a cold process substream. The heat load $q_{l,ls,m,ms,k}$ of every heat transfer match should be distributed to the heat load $qq_{l,ls,m,ms,k,e}$ of the correspond-

ing heat exchanger e . The variable $qq_{l,ls,m,ms,k,e}$ represents the heat load of heat exchanger e between hot process substream l, ls and cold process substream m, ms in temperature interval k . The heat load distribution is achieved by Eqs. 55–66. The number of heat exchangers between a hot process substream and a cold process substream is restricted by Eq. (55), where the parameter E represents the maximum number of heat exchangers between a hot process substream and a cold process substream. The order of heat exchangers is denoted by the variable $zss_{l,ls,m,ms,k}$, which is equal to the sum of $ZZ_{l,ls,m,ms,k}$ from the starting temperature interval of hot stream l to temperature interval k , given by Eq. (56). Then, the binary variable $x_{l,ls,m,ms,k,e}$ is denoted as an indicator to estimate whether $zss_{l,ls,m,ms,k}$ is equal to e , by Eqs. 57–60. When $zss_{l,ls,m,ms,k}$ is equal to e and $Z_{l,ls,m,ms,k}$ is equal to 1, the binary variable $x_{l,ls,m,ms,k,e}$ will be set to 1, which means that the heat transfer in temperature interval k belong to heat exchanger e . Accordingly, $qq_{l,ls,m,ms,k,e}$ will be set to $q_{l,ls,m,ms,k}$ by Eqs. (61), (62). When $zss_{l,ls,m,ms,k}$ is unequal to e or $Z_{l,ls,m,ms,k}$ is equal to 0, $x_{l,ls,m,ms,k,e}$ is set to 0. Therefore, $qq_{l,ls,m,ms,k,e}$ will be 0, given by Eq. (63).

An example is illustrated in Fig. 7, with the table for the value of $Z_{l,ls,m,ms,k}$, $ZZ_{l,ls,m,ms,k}$, $zss_{l,ls,m,ms,k}$, $x_{l,ls,m,ms,k,e}$, and $qq_{l,ls,m,ms,k,e}$. The value

of $z_{l,ls,m,ms,k}$, $zz_{l,ls,m,ms,k}$, and $zsz_{l,ls,m,ms,k}$ are listed in the second, third, and fourth rows respectively, corresponding to the subscript k in the first row. The value of $x_{l,ls,m,ms,k,e}$ and $qq_{l,ls,m,ms,k,e}$ are listed in the fifth and sixth rows separated by a slash, corresponding to the subscript k in the first row and the subscript e in the first column. It is assumed that the maximum number of heat exchangers is two. There are three heat transfer matches in temperature intervals 3, 5, and 6, while the last two (dashed line) can be merged into one heat exchanger. It can be seen that there are three non-zero $z_{l,ls,m,ms,k}$ (heat transfer matches) but only two non-zero $zz_{l,ls,m,ms,k}$ (heat exchangers). The value of variable $zsz_{l,ls,m,ms,k}$ is obtained by Eq. (56). Then, according to the value of $zsz_{l,ls,m,ms,k}$, it is known that the first heat exchanger is located in temperature intervals 3 and 4, while the second one is in temperature intervals 5 and 6. However, there is actually no heat transfer match in temperature interval 4 ($z_{l,ls,m,ms,4}=0$). Thus, the value of $x_{l,ls,m,ms,4,1}$ and $x_{l,ls,m,ms,4,2}$ are both forced to be 1 by Eq. (59). Then the value of $qq_{l,ls,m,ms,4,1}$ and $qq_{l,ls,m,ms,4,2}$ will be 0 owing to Eq. (63). For temperature interval 3, since $zsz_{l,ls,m,ms,3}$ is equal to 1, the value of $x_{l,ls,m,ms,3,2}$ is set to 1 by Eq. (58), while the value of $x_{l,ls,m,ms,3,1}$ can be 0 or 1 given by Eqs. (57) and (58). However, the number of non-zero $x_{l,ls,m,ms,k,e}$ is controlled by Eq. (60). Thus, the value of $x_{l,ls,m,ms,3,1}$ must be 0, and the value of $qq_{l,ls,m,ms,3,1}$ is set to $q_{l,ls,m,ms,3}$ by Eqs. (61) and (62). Similarly, the value of $qq_{l,ls,m,ms,5,2}$ and $qq_{l,ls,m,ms,6,2}$ are equal to $q_{l,ls,m,ms,5}$ and $q_{l,ls,m,ms,6}$ respectively. So far, the heat load ($q_{l,ls,m,ms,3}$) of the heat transfer match in temperature interval 3 is distributed to the heat load ($qq_{l,ls,m,ms,3,1}$) of the first heat exchanger, while the heat loads ($q_{l,ls,m,ms,5}$ and $q_{l,ls,m,ms,6}$) of the heat transfer matches in temperature intervals 5 and 6 are distributed to the heat loads ($qq_{l,ls,m,ms,5,2}$ and $qq_{l,ls,m,ms,6,2}$) of the second heat exchanger.

Note that Eqs. 55–66 become redundant when the formulation of heat exchanger cost is linear. The lower bound, the upper bound, and the specific values for $qq_{l,ls,m,ms,k}$, $zsz_{l,ls,m,ms,k,e}$, and $x_{l,ls,m,ms,k,e}$ are given by Eqs. 64–66.

$$\sum_{k \in \text{TI}} zsz_{l,ls,m,ms,k} \leq E \quad \forall (l, ls) \in \text{HLS}, l \in \text{HP}, (m, ms) \in \text{CMS}, m \in \text{CP} \quad (55)$$

$$zsz_{l,ls,m,ms,k} = \sum_{k' \leq k} z_{l,ls,m,ms,k'} \quad \forall (l, ls) \in \text{HLS}, l \in \text{HP}, (m, ms) \in \text{CMS}, m \in \text{CP}, k \in \text{TI}, ns_l \leq k < ns_m \quad (56)$$

$$zsz_{l,ls,m,ms,k} - e - E \times x_{l,ls,m,ms,k,e} \leq 0 \quad \forall (l, ls) \in \text{HLS}, l \in \text{HP}, (m, ms) \in \text{CMS}, m \in \text{CP}, k \in \text{TI}, ns_l \leq k < ns_m, e \in \text{EX} \quad (57)$$

$$zsz_{l,ls,m,ms,k} - e + E \times x_{l,ls,m,ms,k,e} \geq 0 \quad \forall (l, ls) \in \text{HLS}, l \in \text{HP}, (m, ms) \in \text{CMS}, m \in \text{CP}, k \in \text{TI}, ns_l \leq k < ns_m, e \in \text{EX} \quad (58)$$

$$1 - x_{l,ls,m,ms,k,e} \leq z_{l,ls,m,ms,k} \quad \forall (l, ls) \in \text{HLS}, l \in \text{HP}, (m, ms) \in \text{CMS}, m \in \text{CP}, k \in \text{TI}, ns_l \leq k < ns_m, e \in \text{EX} \quad (59)$$

$$\sum_{k \in \text{TI}} \sum_{e \in \text{EX}} x_{l,ls,m,ms,k,e} = E \times K - \sum_{k \in \text{TI}} z_{l,ls,m,ms,k} \quad \forall (l, ls) \in \text{HLS}, l \in \text{HP}, (m, ms) \in \text{CMS}, m \in \text{CP} \quad (60)$$

$$qq_{l,ls,m,ms,k,e} - q_{l,ls,m,ms,k} - U_{l,m} \times x_{l,ls,m,ms,k,e} \leq 0 \quad \forall (l, ls) \in \text{HLS}, l \in \text{HP}, (m, ms) \in \text{CMS}, m \in \text{CP}, k \in \text{TI}, ns_l \leq k < ns_m, e \in \text{EX} \quad (61)$$

$$qq_{l,ls,m,ms,k,e} - q_{l,ls,m,ms,k} + U_{l,m} \times x_{l,ls,m,ms,k,e} \geq 0 \quad \forall (l, ls) \in \text{HLS}, l \in \text{HP}, (m, ms) \in \text{CMS}, m \in \text{CP}, k \in \text{TI}, ns_l \leq k < ns_m, e \in \text{EX} \quad (62)$$

$$qq_{l,ls,m,ms,k,e} - \Omega_{l,m} \times (1 - x_{l,ls,m,ms,k,e}) \leq 0 \quad \forall (l, ls) \in \text{HLS}, l \in \text{HP}, (m, ms) \in \text{CMS}, m \in \text{CP}, k \in \text{TI}, ns_l \leq k < ns_m, e \in \text{EX} \quad (63)$$

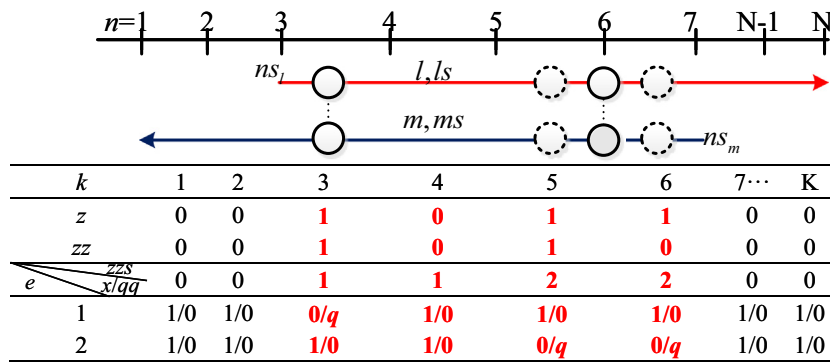
$$0 \leq qq_{l,ls,m,ms,k,e} \leq \Omega_{l,m,k} \quad \forall (l, ls) \in \text{HLS}, (m, ms) \in \text{CMS}, k \in \text{TI}, e \in \text{EX} \quad (64)$$

$$qq_{l,ls,m,ms,k,e} = \begin{cases} 0 & \forall (l, ls) \in \text{HLS}, l \in \text{HP}, (m, ms) \in \text{CMS}, ms \in \text{CP}, k \in \text{TI}, k < ns_l \text{ or } k \geq ns_m, e \in \text{EX} \\ 0 & \forall (l, ls) \notin \text{HLS} \text{ or } (m, ms) \notin \text{CMS}, k \in \text{TI}, e \in \text{EX} \\ 0 & \forall l \in \text{HU} \text{ or } m \in \text{CU}, k \in \text{TI}, e \in \text{EX} \end{cases}$$

$$zsz_{l,ls,m,ms,k} = \begin{cases} 0 & \forall (l, ls) \in \text{HLS}, l \in \text{HP}, (m, ms) \in \text{CMS}, ms \in \text{CP}, k \in \text{TI}, k < ns_l \text{ or } k \geq ns_m \\ 0 & \forall (l, ls) \notin \text{HLS} \text{ or } (m, ms) \notin \text{CMS}, k \in \text{TI} \\ 0 & \forall l \in \text{HU} \text{ or } m \in \text{CU}, k \in \text{TI} \end{cases} \quad (65)$$

$$x_{l,ls,m,ms,k,e} = \begin{cases} 1 & \forall (l, ls) \in \text{HLS}, l \in \text{HP}, (m, ms) \in \text{CMS}, ms \in \text{CP}, k \in \text{TI}, k < ns_l \text{ or } k \geq ns_m, e \in \text{EX} \\ 1 & \forall (l, ls) \notin \text{HLS} \text{ or } (m, ms) \notin \text{CMS}, k \in \text{TI}, e \in \text{EX} \\ 1 & \forall l \in \text{HU} \text{ or } m \in \text{CU}, k \in \text{TI}, e \in \text{EX} \end{cases} \quad (66)$$

Finally, it should be noted that several alternatives that were neglected in the stage-wise superstructure are all included in the proposed model. Above all, both isothermal and non-isothermal mixing are involved in the proposed model, as shown in Fig. 8-1.



$z : z_{l,ls,m,ms,k}, zz : zz_{l,ls,m,ms,k}, zsz : zsz_{l,ls,m,ms,k}, q : q_{l,ls,m,ms,k}, x : x_{l,ls,m,ms,k,e}, qq : qq_{l,ls,m,ms,k,e}$

Fig. 7. An example for heat load counting.

Second, the case of a stream going through two or more heat exchangers in series is available. When the mass flow between sub-streams doesn't exist, a substream can go through two or more heat exchangers in series, as shown in Fig. 8-2. Third, the cases of stream by-pass and recycling are embed in the proposed model. Fig. 8-3 shows one of the possibilities of stream by-pass. Lastly, multiple utilities can be embed in the HEN model. Each utility is available in one temperature interval, which not only can be the last temperature interval or the first temperature interval but also can be the intermediate temperature intervals, as shown in Fig. 8-4.

3.6. Objective function and solution strategy

The objective function of the proposed model is defined as the total annual cost (TAC) including the hot utility cost, the cold utility cost, and the exchanger cost, given by Eq. (67). The hot utility cost (HUC), the cold utility cost (CUC), the sum of exchangers, and the exchanger cost are given by Eqs. (68–71), respectively. Note that the heat exchanger area calculation method in Eq. (71) is different from the literature method. In most of reported papers, the heat exchanger area was obtained according to the total heat load and Chen's approximation of LMTD. In this paper, one heat exchanger maybe placed on more than one temperature intervals. The area of each heat exchanger is obtained by summing the areas among all temperature intervals where the heat exchanger is placed. In each interval, Chen's approximation (Chen, 1987) is used to replace LMTD. Actually, the method adopted in this paper is more accurate than the traditional one in most of reported papers. More details can be found in Appendices B and C.

Objective function:

$$\min TAC = CUC + HUC + EXC \quad (67)$$

Hot utility cost, cold utility cost, sum of exchangers, and exchanger cost:

$$HUC = \sum_{l \in HU} hu_l \times CHU_l \quad (68)$$

$$CUC = \sum_{m \in CU} cu_m \times CCU_m \quad (69)$$

$$EXC = \sum_{(l,ls) \in HLS, l \in HP(m,ms) \in CMS, m \in CP, k \in TI} \sum_{l,ls,ms,k} zz_{l,ls,ms,k} + \sum_{(l,ls) \in HLS, l \in HP(m,ms) \in CMS, m \in CU, m \neq M, k \in TI} \sum_{l,ls,ms,k} zz_{l,ls,ms,k} + \sum_{(l,ls) \in HLS, l \in HU, l \neq L(m,ms) \in CMS, m \in CP, k \in TI} \sum_{l,ls,ms,k} zz_{l,ls,ms,k} + \sum_{m \in CP} zhu_m + \sum_{l \in HP} zcu_l \quad (70)$$

$$EXC = FC \times EXS$$

$$+A \times \sum_{(l,ls) \in HLS, l \in HP(m,ms) \in CMS, m \in CP, e \in EX} \left(\sum_{ns_l \leq k < ns_m} \frac{qq_{l,ls,ms,k,e}}{U_{l,m} \times dt_{l,ls,ms,k}} \right)^B + A \times \sum_{(l,ls) \in HLS, l \in HP(m,ms) \in CMS, m \in CU, m \neq M} \left(\sum_{ns_l \leq k, ne_m \leq k < ns_m} \frac{q_{l,ls,ms,k}}{U_{l,m} \times dt_{l,ls,ms,k}} \right)^B + A \times \sum_{(l,ls) \in HLS, l \in HU, l \neq L(m,ms) \in CMS, m \in CP} \left(\sum_{ns_l \leq k < ne_l, k < ns_m} \frac{q_{l,ls,ms,k}}{U_{l,m} \times dt_{l,ls,ms,k}} \right)^B + A \times \left(\sum_{l \in HP} \left(\frac{\sum_{ls \in LS, ms \in MS} q_{l,ls,ms,k}}{U_{l,m} \times dtcu_l} \right)^B \right) + A \times \left(\sum_{m \in CP} \left(\frac{\sum_{ls \in LS, ms \in MS} q_{l,ls,ms,1}}{U_{l,m} \times dthum} \right)^B \right) \quad (71)$$

The proposed model of HEN is formulated as a MINLP problem, modeled in GAMS 23.3 (General Algebraic Modeling System) on a PC machine (3.2 GHz, 4 G RAM). Note that the nonlinear items of the proposed model only appear in the objective function, similar to the stage-wise superstructure (Yee and Grossmann, 1990), thus the MINLP is robust and can be solved efficiently. Besides, the systematic comparison in Escobar and Trierweiler (2013) found that the Branch and Bound algorithm of SBB/GAMS (Bussieck and Drud, 2001) and the Outer Approximation algorithm of DICOPT/GAMS (Viswanathan and Grossmann, 1990) were more suitable for this kind of MINLP problems. Therefore, we choose the DICOPT solver for the proposed model. Inside DICOPT solver, COINOPT is adopted as NLP solver, while CPLEX is adopted as MILP solver. Note that, DICOPT solver assumes the lower bounds as the default initial values.

4. Case studies

In this section, six examples from the published literature are solved to illustrate the effectiveness of the proposed model. The relevant data for the examples 1–6 (Barbaro and Bagajewicz, 2005; Na et al., 2015; Yee and Grossmann, 1990) are given in Tables 2–4 respectively. Only one hot utility and one cold utility are available for the examples 1–4, while multiple utilities are available for the examples 5 and 6. It should be noted that all reported results are recalculated using the accurate LMTD, including the obtained results of this paper. Besides, the model statistics and computational time are listed in Table 5.

4.1. Example 1

The first example is from Barbaro and Bagajewicz (Barbaro and Bagajewicz, 2005), consisting of two hot and two cold process

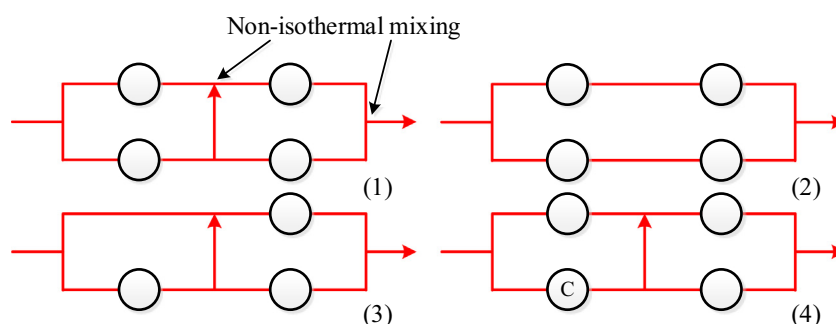


Fig. 8. Network configurations in the proposed model.

Table 2

Data for examples 1,2 (Barbaro and Bagajewicz, 2005).

Stream	T _{in} (°C)	T _{out} (°C)	F (t·h ⁻¹)
<i>Example 1</i>			
H1	175.0	45.0	10.0
H2	125.0	65.0	40.0
HU	180.0	179.0	–
C1	20.0	155.0	20.0
C2	40.0	112.0	15.0
CU	15.0	25.0	–
<i>Example 2</i>			
H1	675.0	150.0	15.0
H2	590.0	450.0	11.0
H3	540.0	115.0	4.5
H4	430.0	345.0	60.0
H5	400.0	100.0	12.0
H6	300.0	230.0	125.0
HU	801.0	800.0	–
C1	60.0	710.0	47.0
CU	80.0	140.0	–

$h = 0.2 \text{ MJ} \cdot \text{h}^{-1} \cdot \text{m}^{-2} \cdot ^\circ\text{C}^{-1}$ for all streams.
 $c_p = 1 \text{ kJ} \cdot \text{kg}^{-1} \cdot ^\circ\text{C}^{-1}$ for all streams.
 Linear formula: $5291.9 + 77.79 \text{Area } \$ \cdot \text{year}^{-1}$.

streams, one hot utility, one cold utility. A minimum temperature difference $\Delta T_{\min}$ of 20 °C, fixed hot and cold utility consumptions, and a linear heat exchanger area cost formulation are adopted from the literature (Barbaro and Bagajewicz, 2005) so that the emphasis lies on the trade-offs between the number and the area of heat exchangers. The temperature intervals are determined: 1 (160 °C) – 2 (155 °C) – 3 (105 °C) – 4 (40 °C) – 5 (20 °C) – 6 (15 °C), according to the starting temperature of all process streams and $\Delta T_{\min}$. Then, the ending temperature 25 °C of cold utility is added into the temperature intervals, so that the cold utility is available in the temperature interval 5 (25 °C– 20 °C). The hot utility can be placed in the temperature interval 1, since there are no other hot process streams available in the temperature interval 1. Besides, the ending temperature of all hot process streams are larger than 20 °C, the temperature level of 15 °C can be deleted. As a result, the final temperature intervals are: 1(160 °C) – 2 (155 °C) – 3 (105 °C) – 4 (40 °C) – 5 (25 °C) – 6 (20 °C). Next, the number of substreams for hot process streams and cold process streams are set to two. The mass flow pattern and the heat flow pattern for example 1

are both illustrated in Fig. 9, where all possible mass flows are present. The circles on substreams indicate all possible heat transfer matches for hot and cold substreams. For example, there are two hot substreams and four cold substreams in the temperature interval 2, thus there are two and four possible heat transfer matches for hot and cold substreams respectively. Since each hot substream can exchange heat with only one cold substream, there are at most two heat transfer matches existing. It is easy to know that there are 12 ($A_4^2 = 4 \times 3 = 12$) combination possibilities for two heat transfer matches in the temperature interval 2. Similarly, there are many combination possibilities for each temperature interval. However, the matching combinations between hot substreams and cold substreams cannot be reflected in Fig. 9, which can be easily obtained by the binary variables in the model. By the proposed model, a HEN with five heat exchangers, one heater, and one cooler is obtained, as shown in Fig. 10-(a), while the result of Barbaro and Bagajewicz (Barbaro and Bagajewicz, 2005) is shown in Fig. 10-(b). The structures of these two networks are similar, especially the right part of the HENs with heat exchangers 3, 4, 5, and cooler. The total area of heat exchangers for Fig. 10-(a) and (b) are 1347.61 m² and 1358.76 m² respectively. The comparisons between the results are listed in Table 6. The total annual cost of 141874.15 \$/yr obtained in this paper is about 0.6% smaller than the one obtained in the literature (Barbaro and Bagajewicz, 2005). The reason why our result is better is that the heat exchanger area items of the objective function in the proposed model are more precise than the one in Barbaro and Bagajewicz (2005).

4.2. Example 2

The second example is originally from Papoulias and Grossmann (1983), then modified by Barbaro and Bagajewicz (2005), consisting of one cold and six hot process streams, one hot utility, one cold utility. Similarly, a minimum temperature difference $\Delta T_{\min}$ of 20 °C, fixed hot and cold utility consumptions, and a linear heat exchanger area cost formulation are adopted. The temperature intervals are determined: 1 (781 °C) – 2 (655 °C) – 3 (570 °C) – 4 (520 °C) – 5 (410 °C) – 6 (380 °C) – 7 (280 °C) – 8 (80 °C) – 9 (60 °C). Similarly, the temperature of the temperature levels 8 and 9 are changed to 140 °C and 80 °C respectively so that the cold utility is available in the temperature interval 8 (140 °C –

Table 3

Data for examples 3,4 (Yee and Grossmann, 1990).

Stream	T _{in} (K)	T _{out} (K)	F _{cp} (kW·K ⁻¹)	Cost (\$·kW ⁻¹ ·year ⁻¹)
<i>Example 3</i>				
H1	443	333	30	–
H2	423	303	15	–
C1	293	408	20	–
C2	353	413	40	–
S1	450	450	–	80
W1	293	313	–	20
$U = 0.8 \text{ kW} \cdot \text{m}^{-2} \cdot \text{K}^{-1}$ for all matches except ones involving steam $U = 1.2 \text{ kW} \cdot \text{m}^{-2} \cdot \text{K}^{-1}$ for all matches involving steam. Annual cost = $1000 \times [\text{area}(\text{m}^2)]^{0.6}$ for all exchangers except heaters. Annual cost = $1200 \times [\text{area}(\text{m}^2)]^{0.6}$ for heaters.				
<i>Example 4</i>				
H1	500	320	6	–
H2	480	380	4	–
H3	460	360	6	–
H4	380	360	20	–
H5	380	320	12	–
C1	290	660	18	–
S1	700	700	–	140
W1	300	320	–	10
$U = 1 \text{ kW} \cdot \text{m}^{-2} \cdot \text{K}^{-1}$ for all matches Annual cost = $1200 \times [\text{area}(\text{m}^2)]^{0.6}$ for all exchangers.				

Table 4

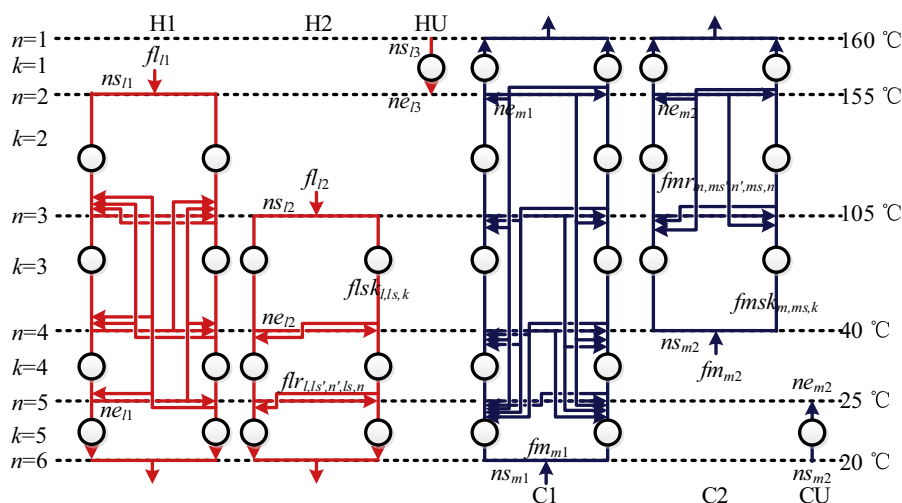
Data for examples 5,6 (Na et al., 2015).

Stream	T_{in} (°C)	T_{out} (°C)	F_{cp} (kW·°C ⁻¹)	h (kW·m ² ·°C ⁻¹)	Cost (\$·kW ⁻¹ ·year ⁻¹)
Example 5					
H1	155	85	150	0.5	–
H2	230	40	85	0.5	–
C1	115	210	140	0.5	–
C2	50	180	55	0.5	–
C3	60	175	60	0.5	–
HP steam	255	254	–	0.5	70
MP steam	205	204	–	0.5	50
LP steam	150	149	–	0.5	20
Cold water	30	40	–	0.5	10
Air cooling	40	65	–	0.5	5
Exchanger capital cost = $1300+1000(\text{area})^{0.83}$, Annualization factor = 0.322.					
Example 6					
H1	600	450	5	0.5	–
H2	250	30	30	0.5	–
C1	50	650	30	0.5	–
Hot oil	330	300	–	5	10
LP steam	400	400	–	5	45
MP steam	630	630	–	5	150
HP steam	680	680	–	5	200
Chilled water	20	25	–	1	20
Stream/Cold utility/Hot utility exchanger capital cost = $500(\text{area})/500(\text{area})/100(\text{area})$.					

Table 5

Model statistics for examples 1–6.

	Single variables	Discrete variables	Single equations	Non-zero elements	Computational time (s)
Example 1	313	140	602	1733	5.58
Ref ^a	381	61	1012	3202	/
Example 2	551	271	1113	3333	8.62
Ref ^a	1106	147	2620	9348	/
Example 3	515	232	1245	3401	13.69
Example 4	1033	429	2138	6688	43.52
Example 5	431	193	934	3482	128.88
Example 6	96	45	204	403	1.28

^a Barbaro and Bagajewicz (2005).**Fig. 9.** Mass flow pattern and heat flow pattern for example 1.

80 °C). The hot utility and cold utility are available in the temperature intervals 1 and 8 respectively. The number of substreams for hot process streams and cold process streams are set to one and three respectively. Besides, the recycled streams are forbidden in this example. The maximum number of heat exchangers is set to ten, including heaters and coolers. The optimized HEN with eight

heat exchangers, one heater, and one cooler is shown in Fig. 11. The example was also solved by the transshipment model (Papoulias and Grossmann, 1983), without considering the heat exchanger cost. Besides, Barbaro and Bagajewicz (2005) solved this example by a one step MILP model, which adopted a formulation to estimate the heat exchanger area. The comparisons among two

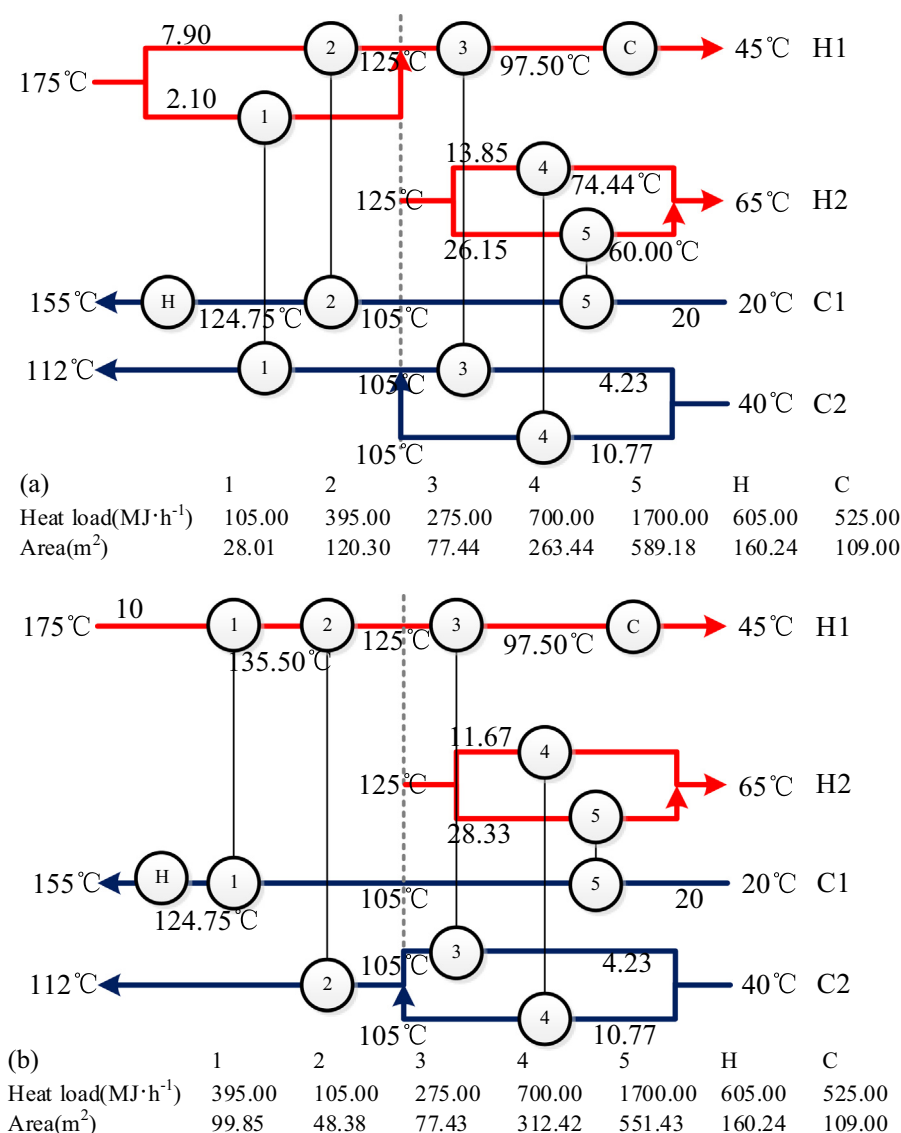


Fig. 10. Heat exchanger network for example 1.

Table 6
Solution comparisons for example 1.

Example 1	Equipments	Heat exchangers area (m ²)	Heat exchangers cost (\$·year ⁻¹)
Ref ^a	7	1358.76	142741.75
This paper	7	1347.61	141874.15

^a Barbaro and Bagajewicz (2005).

reported results (Barbaro and Bagajewicz, 2005; Papoulias and Grossmann, 1983) and the obtained result is listed in Table 7. It shows that the total annual cost of 437121.52 \$/yr is 6.9% and 2.6% smaller than the ones in Papoulias and Grossmann (1983) and Barbaro and Bagajewicz (2005) respectively. The reason for better results is that the exchanger area cost is considered precisely in the proposed model, while it is not considered in Papoulias and Grossmann (1983) and considered approximately in Barbaro and Bagajewicz (2005). As is mentioned before, the model proposed by Barbaro and Bagajewicz (2005) is also a transshipment type HEN model. From Table 5, we can see that the proposed model is more concise than the MILP model of Barbaro and Bagajewicz (2005) in the number of continuous variables and

equations, although the discrete variable number of our model is relatively larger than the reported one.

4.3. Example 3

This example is from Yee and Grossmann (1990), aiming to illustrate the scenario when a substream goes through two or more heat exchangers in series. This example has two hot process streams, two cold process streams, one hot utility (steam), and one cold utility (water). The minimum temperature difference $\Delta T_{\min}$ is assumed to be 2 K. According to the starting temperature of all streams and $\Delta T_{\min}$, the temperature intervals are determined: 1 (448 K) – 2 (441 K) – 3 (421 K) – 4 (353 K) – 5 (293 K). Since the biggest ending temperature of cold streams is smaller than the temperature level 3, the temperature level 1 is redundant. The hot utility can be placed in the temperature interval 441 K – 421 K. Then, the ending temperature of cold utility is added, so that the cold utility is available in the last temperature interval. The final temperature intervals are: 1 (441 K) – 2 (421 K) – 3 (353 K) – 4 (313 K) – 5 (293 K). The number of substreams for hot process streams and cold process streams are set to two. In order to compare with the result (Fig. 12-a) of Yee and Grossmann (1990), the

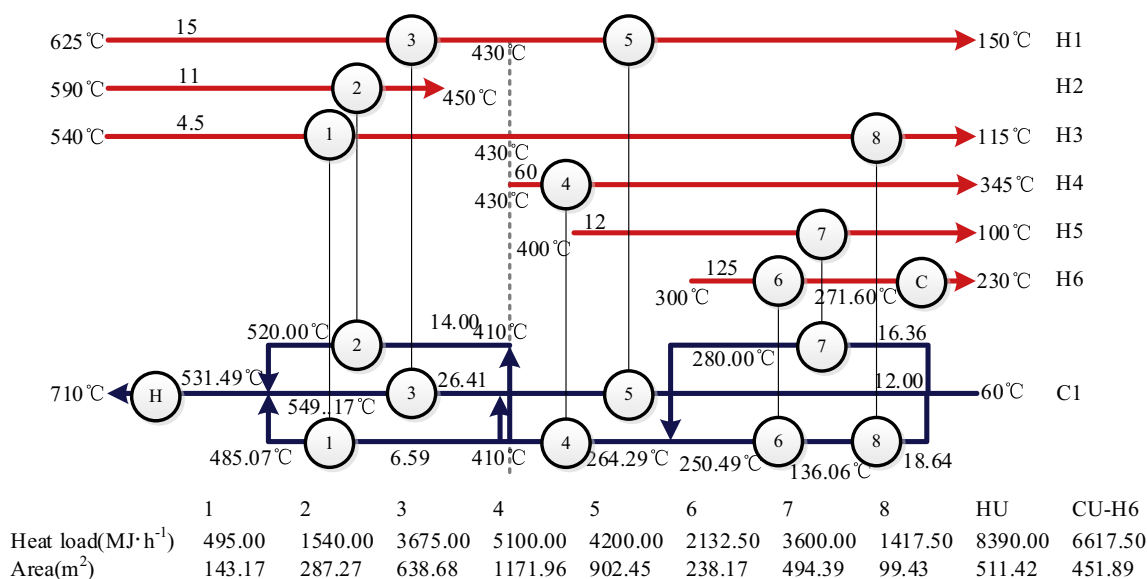


Fig. 11. Heat exchanger network for example 2.

Table 7
Solution comparisons for example 2.

Example 2	Equipment	Heat exchangers area (m ²)	Heat exchangers cost (\$·year ⁻¹)
Ref ^a	10	5355.55	469573.42
Ref ^b	10	5087.07	448682.33
This paper	10	4938.82	437121.52

^a Papoulias and Grossmann (1983).

^b Barbaro and Bagajewicz (2005).

hot and cold utility consumptions are fixed at the same values as Yee and Grossmann (1990). Besides, the maximum number of heat exchangers is set to five. Using the proposed model, a HEN with four heat exchangers and one cooler is obtained in 13.69 s, as shown in Fig. 12-b. Note that, one substream of cold stream C1 goes through two heat exchangers in series with hot stream H1. It is well-known that the stage-wise superstructure can't yield

such configuration. The results comparisons are listed in Table 8. The obtained total annual cost (TAC) is 76762.71\$/yr, 4.4% lower than the one in Yee and Grossmann (1990).

4.4. Example 4

This example is also from Yee and Grossmann (1990) with five hot and one cold process streams, and one hot and one cold utility. In this example, the minimum temperature difference $\Delta T_{\min}$ is assumed to be 5 K. Besides, the starting and ending temperatures of hot and cold process streams are all adopted to construct the temperature intervals: 1 (660 K) – 2 (495 K) – 3 (475 K) – 4 (455 K) – 5 (375 K) – 6 (355 K) – 7 (315 K) – 8 (290 K). The hot utility can be placed in the temperature interval 1, since there is no hot process streams available in the temperature interval 1. Then the temperature levels 7 and 8 are changed to 320 K and 300 K respectively, so that the cold utility can be placed in the last temperature

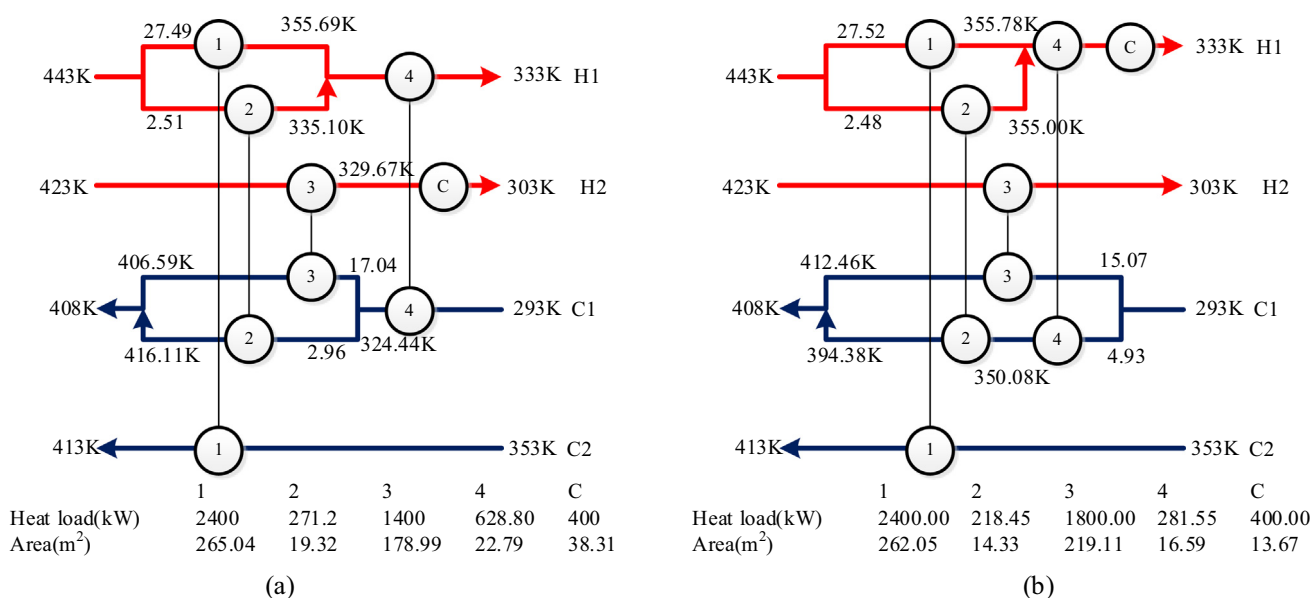


Fig. 12. Heat exchanger network for example 3.

Table 8
Solution comparisons for example 3.

	Equipment	EXC (\$/yr)	UC (\$/yr)	TAC (\$/yr)
Ref ^a	5	72625.63	8000	80625.63
Ref ^b	5	72266.84	8000	80266.84
This paper	5	68762.71	8000	76762.71

^a Toffolo (2009).

^b Yee and Grossmann (1990).

interval 7. Although the starting temperature of cold process stream C1 is smaller than the temperature level 8, there is no need to add another temperature level of 290 K because the ending temperature of all hot process streams are all higher than 300 K. The number of substreams of hot process streams is set to one, while it is set to five for cold process stream as the sum of hot process streams. The recycled streams are forbidden in temperature levels 1–4. The maximum hot utility and cold utility consumption are set to 3580 kW and 120 kW respectively, while the maximum number of heat exchangers is set to seven. The proposed model yields a HEN with the TAC of 571323.01 \$/yr, as shown in Fig. 13. In the obtained HEN, two recycling streams flow from the temperature level 355 K to the temperature level 315 K, mixed with the substreams of 290 K. Note that, this kind of structure cannot be obtained by the stage-wise superstructure (Yee and Grossmann, 1990). Table 9 shows the comparisons among the one obtained by the stage-wise superstructure (Yee and Grossmann, 1990), the one obtained by a multistage superstructure with cross flows (Huang and Karimi, 2013), and the one obtained in this paper. The obtained TAC is 0.9% smaller than the one in Yee and Grossmann (1990) and 0.2% larger than the one in Huang and Karimi (2013). We recognize the proposed model cannot yield the same result of Huang and Karimi (2013) unless the temperature levels set to specific values. Nevertheless, the obtained result is only slightly worse. Besides, the model of Huang and Karimi (2013) is too complicated to be solved by DICOPT/GAMS. Instead, they solved it by BARON/GAMS with a maximum solution time limit (48 h of CPU time). Nevertheless, the proposed model is more robust with all linear constraints, which can be solved by DICOPT/GAMS quickly in 43.52 s for example 4.

Table 9
Solution comparisons for example 4.

	Equipment	EXC (\$/yr)	UC (\$/yr)	TAC (\$/yr)
Ref ^a	7	59786.18	516860.20	576366.38
Ref ^b	7	69289.21	501110.00	570399.21
This paper	7	68923.01	502400.00	571323.01

^a Yee and Grossmann (1990).

^b Huang and Karimi (2013).

4.5. Example 5

This is a representative problem with multiple utilities from Shenoy et al. (1998), which has been solved in many HEN models (Isafiade and Fraser, 2008; Na et al., 2015; Ponce-Ortega et al., 2010). It has two hot and three cold process streams, three hot utilities (HP steam, MP steam, LP steam), and two cold utilities (air cooling, cool water). The stream data and the cost function are given in Table 4. The minimum temperature difference $\Delta T_{\min}$ is assumed to be 5 °C. According to the starting temperature of all streams and the $\Delta T_{\min}$, the temperature intervals are determined: 1 (250 °C) – 2 (225 °C) – 3 (200 °C) – 4 (150 °C) – 5 (145 °C) – 6 (115 °C) – 7 (65 °C) – 8 (40 °C) – 9 (30 °C). Note that the starting temperature of cold process streams C2 and C3 are not chose to construct the temperature intervals so that cold utilities air cooling and cool water are available in the temperature intervals 7 and 8 respectively. The starting temperature levels for C2 and C3 are set in the temperature level 7. The hot utilities of HP steam, MP steam, and LP steam are available in the temperature interval 1, 3, and 4 respectively. The number of substreams is set to one for cold process streams and cold utilities, and two for hot process streams and hot utilities. The maximum hot utility and cold utility consumption are set to 8400 kW and 7700 kW respectively, while the maximum number of heat exchangers is set to eight. Fig. 14 shows the optimized HEN, with TAC of 1116790.0 \$/yr. Table 10 shows comparisons among the obtained result and the reported results. The reported TAC of Na et al. (2015) and Huang and Karimi (2013) are 1120619 \$/yr and 1115868 \$/yr respectively. The result obtained by the proposed model is 1118310.62 \$/yr.

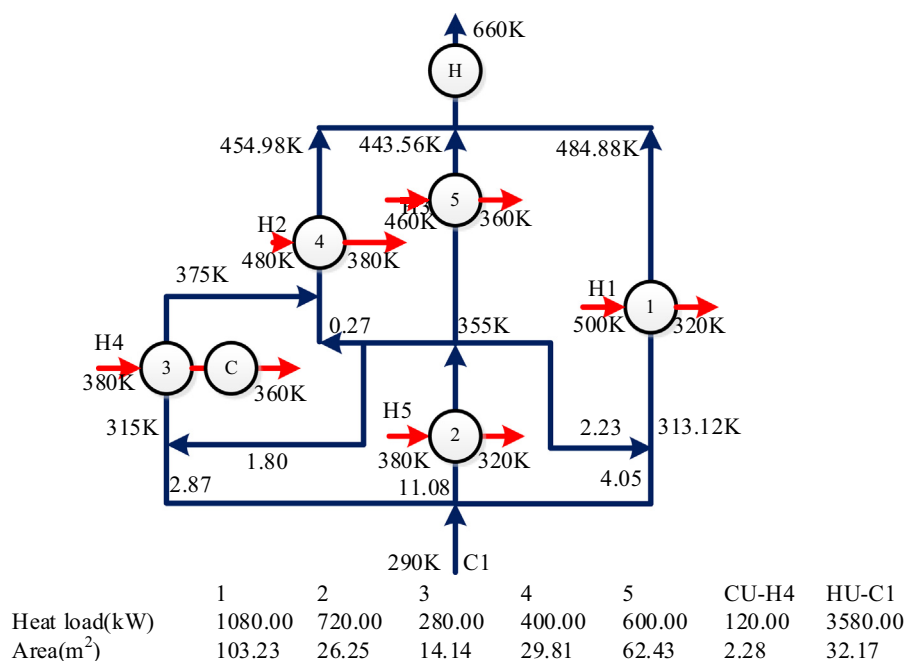


Fig. 13. Heat exchanger network for example 4.

Chen's LMTD approximation (Chen, 1987) and Paterson's approximation (Paterson, 1984) were adopted in Na et al. (2015) and Huang and Karimi (2013) respectively. As is well known, it will overestimate heat exchanger cost by Chen's approximation (Chen, 1987), while Paterson's approximations (Paterson, 1984) underestimate heat exchanger cost. Although the reported solution of Huang and Karimi (2013) is best, it shows that the proposed model gives a slightly better solution after recalculated by the accurate LMTD. Note that, the main differences among the HEN structures of reported solutions (Huang and Karimi, 2013; Na et al., 2015; Ponce-Ortega et al., 2010) and Fig. 14 are the location of cooler C2 and the split ratio of hot process stream H1. The cooler C2 is implemented on the mixing stream of two substreams in Na et al. and Ponce-Ortega et al. (Huang and Karimi, 2013; Na et al., 2015; Ponce-Ortega et al., 2010), while it is implemented on one of the substreams in Huang and Karimi (Huang and Karimi, 2013; Na et al., 2015; Ponce-Ortega et al., 2010) and this paper. The result comparisons reveal that our structure is a better choice for this case. Comparing our result with the one in Huang and Karimi (Huang and Karimi, 2013; Na et al., 2015; Ponce-Ortega et al., 2010), the HEN structures are identical. However, since a more accurate method for heat exchanger area calculation is adopted in this paper, the trade-off between the utility consumption, the split ratio of streams, and the heat load of each heat exchanger can be considered more precisely.

4.6. Example 6

This is an example with multiple utility adopted from Na et al. (2015) In the example, there are two hot process stream, one cold

process stream, four hot utilities (HP steam, MP steam, LP steam, Hot oil), and one cold utility (Chilled water). The stream data and the cost function are given in Table 4. The minimum temperature difference $\Delta T_{\min}$ is assumed to be 5 °C. Then, the temperature intervals are determined: 1 (675 °C) – 2 (625 °C) – 3 (595 °C) – 4 (395 °C) – 5 (325 °C) – 6 (245 °C) – 7 (50 °C) – 8 (20 °C). The hot utilities of HP steam, MP steam, LP steam, and hot oil are available in the temperature intervals 1, 2, 4, and 5 respectively, while the cold utilities of chilled water is available in the temperature interval 7. The number of substreams is set to one for hot process streams, hot utility, cold process streams and cold utility. Besides, Chen's LMTD approximation (Chen, 1987) is replaced by the one of Paterson (1984) in this example to keep consistence with Na et al. (2015). Then, the optimized HEN is obtained by running the proposed model with the default initial points, as shown in Fig. 15. The obtained TAC is 1557612.03 \$/yr, including EXC of 2662224.36 \$/yr and UC of 1291388.77 \$/yr. Table 11 compares the results of this paper with the one in Na et al. (2015), including hot utility consumption, cold utility consumption, IC, UC, and TAC. The utility cost is decreased at the cost of increasing the investment cost. The total annual cost is 0.4% better than the reported result.

5. Conclusion

In this paper, a new type of transshipment model is proposed for the heat exchanger network synthesis. Compared to the traditional transshipment models, the proposed model not only consider the heat flow pattern but also include the mass flow pattern of hot and cold streams. As a result, splitting, by-passing

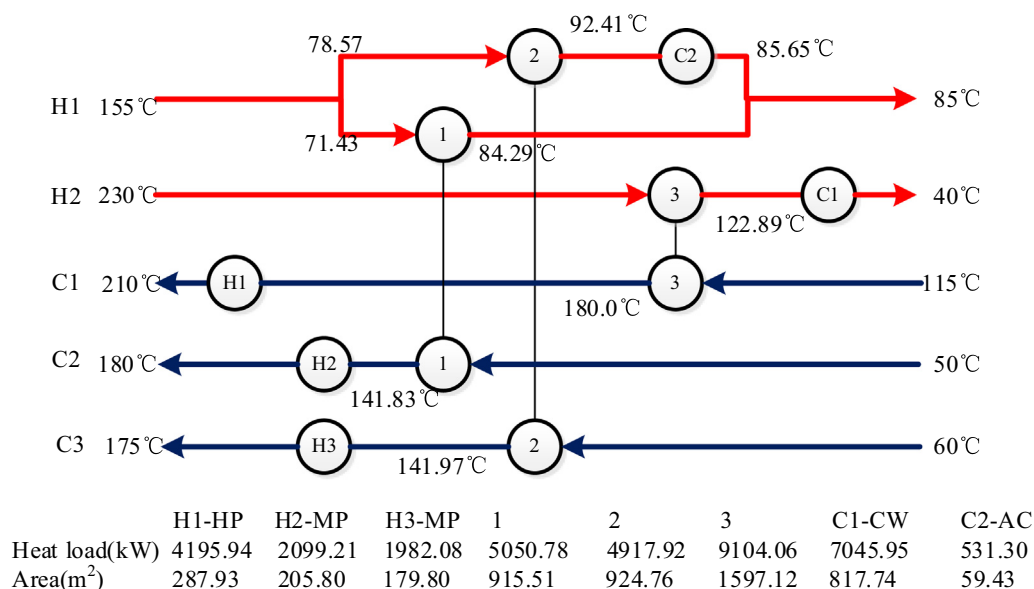


Fig. 14. Heat exchanger network for example 5.

Table 10

Solution comparisons for example 5.

	HP/MP/LP steam (kW)	Air cooling/Cold water (kW)	EXC (\$/yr)	UC (\$/yr)	TAC (\$/yr)
Ref ^a	4885.02/3575.00/0	4160.00/3600.00	577935.56	580301.15	1158236.71
Ref ^b	4186.42/4079.75/0	7036.39/529.69	546754.24	570045.93	1116800.16
Ref ^c	4228.56/4083.11/0	7611.73/0	541262.74	576272.25	1117535.00
This paper	4195.94/4081.29/0	7045.95/531.30	545893.54	570896.50	1116790.05

^a Shenoy et al. (1998).

^b Huang and Karimi (2013).

^c Na et al. (2015).

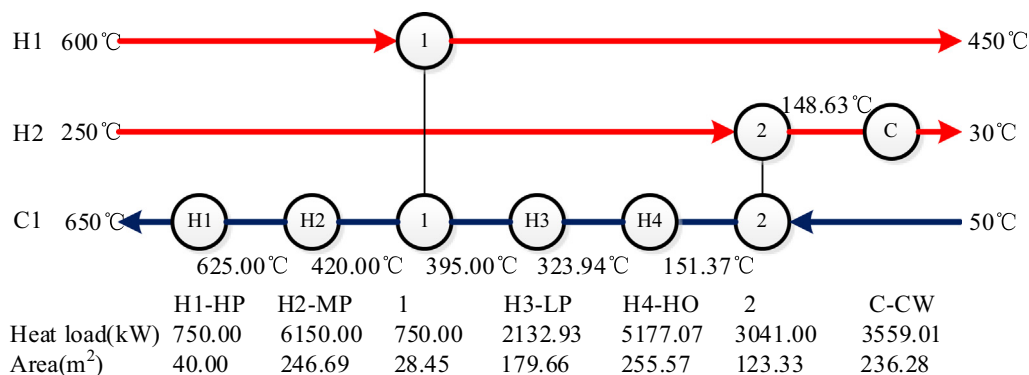


Fig. 15. Heat exchanger network for example 6.

Table 11
Solution comparisons for example 6.

	HP/MP/LP steam/Hot oil (kW)	Chilled water (kW)	EXC (\$/yr)	UC (\$/yr)	TAC (\$/yr)
Ref ^a	752.70/6039.20/2589.80/5402.70	4134.40	254382.46	1309667.66	1564050.11
This paper	750.00/6150.00/2131.39/5177.07	3559.01	266224.36	1291388.77	1557612.03

^a Na et al. (2015).

and recycling of streams as well as isothermal and non-isothermal mixing of streams are all included in the model. Similar to the widely used stage-wise superstructure of Yee and Grossmann (1990), the proposed model is a MINLP problem with linear constraints, which can be solved by DICOPT/GAMS efficiently. Besides, a new method is developed to identify heat exchangers, while the heat exchanger areas can be obtained using a more accurate method by the piecewise calculation. Six examples from literature including two examples with multiple utilities are solved to illustrate the effectiveness and applicability of the proposed model. Due to the richer structure possibilities, the linear constraints as well as the more accurate temperature difference calculation method, the proposed model yields better HEN configurations with lower TAC for most of the illustrated examples.

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Appendix A. Partition of temperature intervals

The specific steps are as follows:

- (1) The minimum temperature difference $\Delta T_{\min}$ is determined. The starting and ending temperatures of hot streams are subtracted by $\Delta T_{\min}$.
- (2) The starting temperature and the ending temperature of hot streams (H) and cold streams (C) are ranked in decreasing order, including process streams (HP and CP) and utilities (HU and CU). Then, the temperature intervals are constructed.

Since the number of intervals is a key factor for the size of the model, there are two rules to adjust the temperature intervals:

- (1) It is known that the ending temperature of streams can't be pinch temperature. Therefore, it is feasible to just choose the starting temperature of hot streams and cold streams to construct the temperature intervals, in order to reduce the number of temperature intervals. Under the circumstances, if there are cold streams whose ending temperatures are bigger than the biggest starting temperature of all hot streams, the biggest ending temperature of cold streams should be added to the temperature intervals. Similarly, if there are hot streams whose ending temperatures are smaller than the smallest starting temperature of all cold streams, the smallest ending temperature of hot streams should be added to the temperature intervals. In a word, the biggest temperature in the temperature intervals must be larger than the ending temperature of all cold streams. The smallest temperature in the temperature intervals must be smaller than the starting temperature of all hot streams.
- (2) If the temperature range of a certain temperature interval is much bigger than the ones of other temperature intervals, the temperature interval can be divided into two or more temperature intervals by adding the ending temperature of streams or the average temperature of this temperature interval. As a result, hot (cold) streams can exchange heat with more than one cold (hot) streams in this temperature range. More network structure possibilities will be included.

After constructing the temperature intervals, the starting temperature level (ns) and the ending temperature level (ne) of hot streams and cold streams should be determined. Since not all the starting and ending temperatures of streams are chosen to construct the temperature intervals, there should be rules to determine ns and ne for hot streams and cold streams. For cold stream m , if the starting/ending temperature is equal to the temperature of a certain temperature level, the temperature level is ns_m/ne_m . If not, the temperature level which is closest to and bigger than the starting/ending temperature will be ns_m/ne_m . Similarly, for hot stream l , if the starting/ending temperature is equal to the temperature of a certain temperature level, the temperature level is

ns_i/ne_i . If not, the temperature level which is closest to and smaller than the starting/ending temperature will be ns_i/ne_i . Note that since the starting and ending temperatures of all streams and the temperature of all temperature levels are constants, it is easy to keep the energy balance for each stream.

Appendix B. Approximation of LMTD

As is well known, the approximation of LMTD by Chen (Chen, 1987) is widely used in HENS problems. From the comparisons between LMTD (θ_{LM}) and Chen's approximation (θ_{CM}) in Table A1, it is known that Chen's approximation (Chen, 1987) underestimate the temperature difference, which will overestimate the heat exchanger area. Besides, the relative error increases when the ratio between θ_2 and θ_1 increases. Parameters θ_2 and θ_1 represent the temperature differences at two ends of the heat exchanger ($\theta_2 \geq \theta_1$). Fig. A1 shows the relative error when the ratio is between 1 and 15.

The heat exchanger area is obtained by the total heat load and the LMTD approximation by Chen (1987) in most of reported papers. In the proposed method, the heat exchanger area is obtained by summing the areas among all intervals where the heat exchanger is placed. Chen's approximation (Chen, 1987) is also adopted for each temperature interval. An example is illustrated in Fig. A2. The heat exchanger covers three temperature intervals, where θ_1 , θ_2 , θ_3 and θ_4 represent the temperature differences between the hot stream and the cold stream in the temperature intervals 1, 2, and 3 respectively. The variables Q_1 , Q_2 , and Q_3 represent the heat load of each heat transfer match, while the variable Q represents the total heat load. The proof procedure is as follows:

Given: Q , Q_1 , Q_2 , Q_3 , U , θ_1 , θ_2 , θ_3 , θ_4

The accurate areas can be obtained:

$$a_L = \frac{Q}{U \frac{\theta_4 - \theta_1}{\ln \frac{\theta_4}{\theta_1}}}, a_{1L} = \frac{Q_1}{U \frac{\theta_2 - \theta_1}{\ln \frac{\theta_2}{\theta_1}}}, a_{2L} = \frac{Q_2}{U \frac{\theta_3 - \theta_2}{\ln \frac{\theta_3}{\theta_2}}}, a_{3L} = \frac{Q_3}{U \frac{\theta_4 - \theta_3}{\ln \frac{\theta_4}{\theta_3}}}$$

where the variables a_{1L} , a_{2L} , and a_{3L} represent the heat exchanger area of the temperature intervals 1, 2, and 3 respectively. The variable a_L represents the total heat exchanger area. Since the accurate LMTD is adopted, the sum of a_{1L} , a_{2L} , and a_{3L} must be equal to a_L .

$$a_L = a_{1L} + a_{2L} + a_{3L}$$

Then, the areas calculated by Chen's approximation (Chen, 1987) are obtained:

$$a_C = \frac{Q}{U[\theta_4 \times \theta_1 \times (\theta_4 + \theta_1)/2]^{\frac{1}{3}}}, a_{1C} = \frac{Q_1}{U[\theta_2 \times \theta_1 \times (\theta_2 + \theta_1)/2]^{\frac{1}{3}}}, a_{2C} = \frac{Q_2}{U[\theta_3 \times \theta_2 \times (\theta_3 + \theta_2)/2]^{\frac{1}{3}}}, a_{3C} = \frac{Q_3}{U[\theta_4 \times \theta_3 \times (\theta_4 + \theta_3)/2]^{\frac{1}{3}}}$$

Defind: $R = \frac{|\theta_{CM} - \theta_{LM}|}{\theta_{LM}}$, $0 < R < 1$

$$\therefore \theta_{CM} = (1 - R)\theta_{LM} \therefore a_{CM} = \frac{a_{LM}}{1 - R}$$

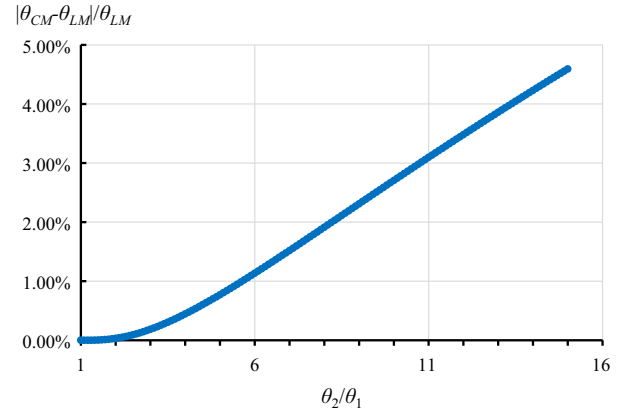


Fig. A1. The relation between the relative error and the ratio of θ_2 and θ_1 .

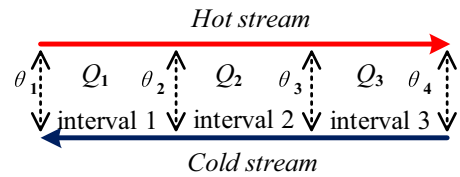


Fig. A2. An example of one heat exchanger with three temperature intervals.

$$a_C = \frac{Q}{U[\theta_4 \times \theta_1 \times (\theta_4 + \theta_1)/2]^{\frac{1}{3}}} = \frac{Q}{U(1-R_{14})^{\frac{\theta_4 - \theta_1}{\ln \frac{\theta_4}{\theta_1}}}} = \frac{a_L}{1-R_{14}} = \frac{a_{1L} + a_{2L} + a_{3L}}{(1-R_{14})}$$

$$\therefore a_{1C} = \frac{Q_1}{U[\theta_2 \times \theta_1 \times (\theta_2 + \theta_1)/2]^{\frac{1}{3}}} = \frac{a_{1L}}{(1-R_{12})}$$

$$a_{2C} = \frac{Q_2}{U[\theta_3 \times \theta_2 \times (\theta_3 + \theta_2)/2]^{\frac{1}{3}}} = \frac{a_{2L}}{(1-R_{23})}$$

$$a_{3C} = \frac{Q_3}{U[\theta_4 \times \theta_3 \times (\theta_4 + \theta_3)/2]^{\frac{1}{3}}} = \frac{a_{3L}}{(1-R_{34})}$$

$$\therefore \frac{\theta_4}{\theta_1} > \frac{\theta_2}{\theta_1}, \frac{\theta_3}{\theta_2}, \frac{\theta_4}{\theta_3}$$

$$\therefore R_{14} > R_{12}, R_{23}, R_{34}$$

$$\therefore a_{1L} < \frac{a_{1L}}{(1-R_{12})} < \frac{a_{1L}}{(1-R_{14})}, a_{2L} < \frac{a_{2L}}{(1-R_{23})} < \frac{a_{2L}}{(1-R_{14})},$$

$$a_{3L} < \frac{a_{3L}}{(1-R_{34})} < \frac{a_{3L}}{(1-R_{14})}$$

$$\therefore a_{1L} + a_{2L} + a_{3L} < \frac{a_{1L}}{(1-R_{12})} + \frac{a_{2L}}{(1-R_{23})} + \frac{a_{3L}}{(1-R_{34})} < \frac{a_{1L}}{(1-R_{14})} + \frac{a_{2L}}{(1-R_{14})} + \frac{a_{3L}}{(1-R_{14})}$$

$$\therefore a_{1L} + a_{2L} + a_{3L} < a_{1C} + a_{2C} + a_{3C} < \frac{a_{1L} + a_{2L} + a_{3L}}{(1-R_{14})}$$

$$\therefore a_L < a_{1C} + a_{2C} + a_{3C} < a_C$$

Table A1

Comparisons between LMTD and Chen's approximation.

$\theta_1 / ^\circ\text{C}$	10	10	10	10	10	10	10	10
$\theta_2 / ^\circ\text{C}$	10	30	50	70	90	110	130	150
θ_2/θ_1	1	3	5	7	9	11	13	15
θ_{LM}	10	18.21	24.85	30.83	36.41	41.70	46.78	51.70
θ_{CM}	10	18.17	24.66	30.37	35.57	40.41	44.98	49.32
$ (\theta_{CM} - \theta_{LM}) /\theta_{LM}$	0	0.18%	0.77%	1.52%	2.31%	3.10%	3.86%	4.59%

It proves that the area ($a_{1C}+a_{2C}+a_{3C}$) obtained by the proposed method is larger than the accurate area a_L ($a_L=a_{1L}+a_{2L}+a_{3L}$) and smaller than the value (a_C) obtained by the common method. Therefore, the method proposed in this paper is more accurate than the common one in most of reported papers.

Appendix C. Comments on approximations for LMTD

In this section, we compare two reported approximations (Chen, 1987) and the method proposed in this paper. As shown in Fig. A3, the heat exchanger area is calculated by four different cases. The modification of Paterson (1984) and the modification of Underwood (1970) by Chen (1987) are adopted in the first case and second case respectively. In the third and fourth cases, the heat exchanger is treated as two and three heat transfer matches respectively. The total heat load is averagely divided into two and three parts respectively. The modification of Paterson (1984) by Chen (1987) is adopted in these two cases. Then, the relative errors between the obtained heat exchanger area and the exact area (a_{LM}) obtained by LMTD for four cases are shown in Fig. A4. So far, the modification of Underwood (1970) by Chen (1987) is the most accurate reported approximation for LMTD. It is clear that the relative error of case 1 is larger than the one of case 2, as shown in Fig. A4. Besides, the relative error of case 1 is the largest among three cases 1, 3, and 4, where same approximation (Chen, 1987) is adopted. The relative error is in the range of 0~4.81%, 0~1.50%, and 0~0.66% for cases 1, 3, and 4 respectively, when the ratio between θ_2 and θ_1 increase from 1 to 15. The relative errors for case 3 and case 4 are much less than the one for case 1. It shows that the piecewise calculation has a significant effect on improving the accuracy. For clear comparison, cases 2, 3, and 4 are shown in

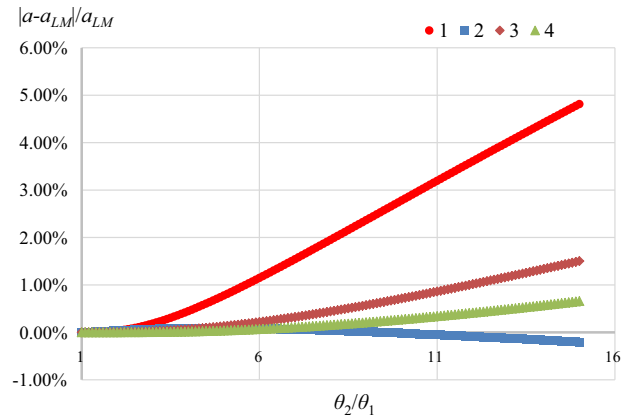


Fig. A4. The relative error of the heat exchanger area for cases 1, 2, 3, and 4.

Fig. A5. For case 3, when the ratio between θ_2 and θ_1 is smaller than 4.6, the relative error is smaller than the one for case 2. For, case 4, when the ratio between θ_2 and θ_1 is smaller than 6.3, the relative error is smaller than the one for case 2. Although the approximation adopted in case 2 is more accurate than the one in cases 3 and 4, the results of cases 3 and 4 can be better than case 2, owing to the proposed method by the piecewise calculation. Note that, when a more accurate approximation or a partition method with more segments is adopted in case 3 and case 4, the results must be more accurate. It is concluded that the method proposed in this paper by the piecewise calculation can improve the accuracy when using an approximation of LMTD. Besides, the proposed method is suitable for other HEN models, such as the stage-wise superstructure of Yee and Grossmann (1990).

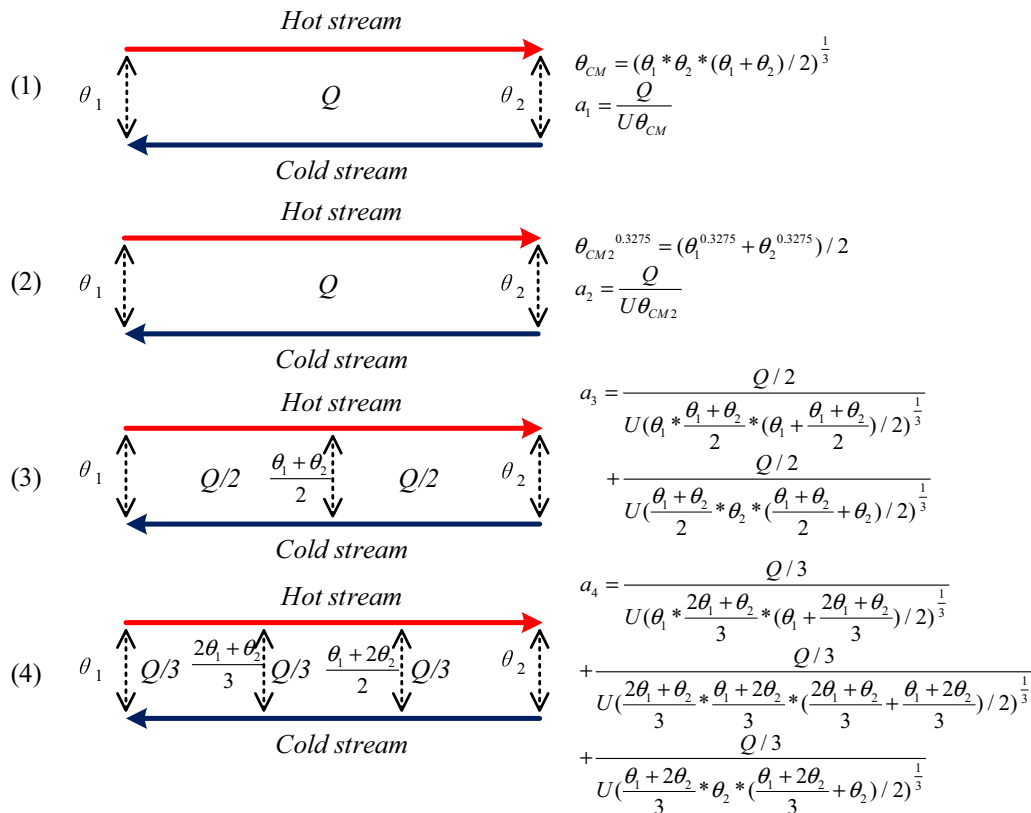


Fig. A3. Four cases to calculate the heat exchanger area.

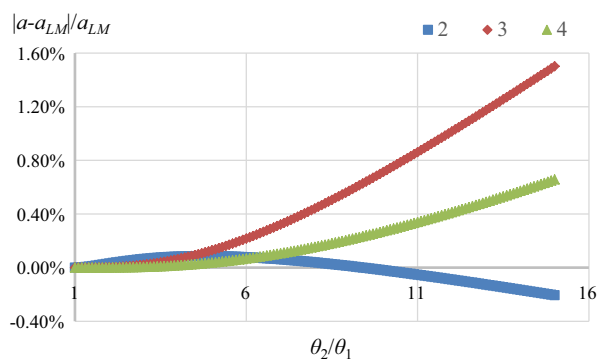


Fig. A5. The relative error of the heat exchanger area for cases 2, 3, and 4.

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